SMART PHARMACY INVENTORY & EXPIRY ALERT SYSTEM

A PROJECT REPORT

Submitted by

POOJA D

(2116240701385)

in partial fulfilment of the End Semester Examination for the course

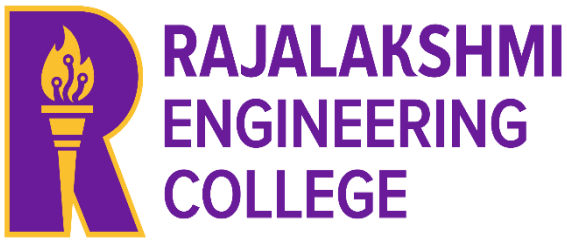
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "SMART PHARMACY INVENTORY & EXPIRY ALERT SYSTEM" is the bonafide work of POOJA D (240701385), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

DECLARATION

I, POOJA D (240701385), hereby declare that the project report titled "SMART PHARMACY INVENTORY & EXPIRY ALERT SYSTEM", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

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Date:

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POOJA D

ABSTRACT

Pharmacies must track medicines batch by batch so that expired stock is never dispensed, stock quantities stay accurate and shortages are noticed before they affect patients. When medicine details, expiry dates, purchases and issues are kept in separate registers or spreadsheets, staff have no reliable view of what is in stock, which batches are about to expire and which medicines need reordering. This project developed the Smart Pharmacy Inventory & Expiry Alert System, a medicine inventory and expiry management application, to bring these activities together in one application and give pharmacy staff and managers a consistent view of stock. The system was built in Zoho Creator using eight connected forms: Medicine, Create the Supplier form, Supplier, Purchase form, Batch Inventory form, Stock Issue, Reorder Request and Stock Adjustment. Lookup fields link purchases, batches, issues, adjustments and reorder requests to the medicines and suppliers they refer to. Deluge scripts and form workflows reject batches whose expiry date is in the past, automatically reduce a batch's available quantity when stock is issued, send a low-stock alert e-mail when the available quantity falls to 10 units or fewer, and send a notification e-mail when a reorder request is submitted. Ten list reports, including an Urgent Reorders report, and two dashboard pages with KPI cards and charts present stock, batch status and pending reorders. Role-based access is provided through Pharmacist Access, Inventory Staff Access and Pharmacy Admin Access permission profiles, a role hierarchy and a sharing rule that gives pharmacists read access to batch records. The application is available through Zoho Creator's web interface and mobile app and uses only built-in forms, reports, workflow, notification and permission capabilities, giving pharmacy staff a common place to record, monitor and replenish medicine stock.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Pharmacy Inventory Management; Expiry Alert; Workflow Automation; Role-Based Access Control.

PROJECT AT A GLANCE

Attribute	Details
Project Title	Smart Pharmacy Inventory & Expiry Alert System
Application Domain	Healthcare – pharmacy inventory, batch and expiry management
Student Name & Register No.	Pooja D – 240701385
Zoho Creator Edition	Zoho Creator (free trial edition)
Application Name	Smart Pharmacy Inventory & Expiry Alert System (icon text “SPI”)
Published / Live URL	https://creatorapp.zoho.in/pooja.d.2024.cse_rajalaks64/smart-pharmacy-inventory-expiry-alert-system/#Form:Stock_Adjustment
Demo Credentials (per role)	Administrator: application owner account (pooja.d.2024.cse@rajalakshmi.edu.in) | Pharmacist, Inventory Staff and Pharmacy Admin permission profiles are defined
Number of Forms / Subforms	8 / 0
Number of Reports (by type)	List: 10 (8 default + 2 custom) Kanban: 0 Calendar: 0 Pivot/Chart: 0
Number of Pages / Dashboards	2 (Pharmacy_Dashboard, Pharmacy Dashboard 2)
Number of Workflows	Form: 4 Schedule: 0 Approval: 0 Blueprint: 0 Batch: 0
Deluge Functions / Custom Actions	2 Deluge scripts (expiry validation, automatic stock update) and 2 send-mail actions; 0 stored functions
Integrations	Zoho Mail (sendmail task)
Roles & Profiles Defined	Administrator; permission profiles: Pharmacist Access, Inventory Staff Access, Pharmacy Admin Access; roles: CEO, Pharmacist, Inventory Staff, Pharmacy Admin; 1 sharing rule
Environments Used	Single application workspace; separate Stage and Production environments were not configured
Date of Production Deployment	Not separately deployed; live application URL used for demonstration

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
CRUD	Create, Read, Update, Delete
DS	Deluge Script
ER	Entity–Relationship
FR	Functional Requirement
GST	Goods and Services Tax
KPI	Key Performance Indicator
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
REST	Representational State Transfer
SPI	Smart Pharmacy Inventory
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Pharmacies in hospitals, colleges, clinics and retail outlets stock hundreds of medicines such as tablets, capsules, syrups, injections and ointments. Every medicine arrives in batches, and each batch carries its own manufacturing date, expiry date and quantity. A pharmacy must therefore know not only how much of a medicine it holds, but also which batch the stock belongs to, where it is stored, when it will expire and when it needs to be reordered. Dispensing expired medicine is a patient-safety risk and a regulatory violation, while running out of a critical medicine disrupts patient care.

In many small and medium pharmacies this information is still maintained in purchase registers, stock books, spreadsheets and telephone calls to suppliers. The Smart Pharmacy Inventory & Expiry Alert System replaces these scattered records with a single Zoho Creator application that covers medicines, suppliers, purchases, batch inventory, stock issues, reorder requests and stock adjustments, supported by automatic stock updates, e-mail alerts and a dashboard.

1.2 Problem Statement

Medicine stock, batch details, expiry dates and supplier purchases are recorded manually in separate registers and files, so expired or near-expiry stock is noticed late, stock quantities are not updated when medicines are issued, shortages are discovered only after they occur, and reorder decisions are not tracked. The specific pain points are:

- **Expiry risk:** expiry dates are checked by visual inspection, so expired medicines can remain on shelves or be entered into stock.
- **Inaccurate stock levels:** quantities are not reduced immediately when medicines are issued, so recorded and physical stock differ.
- **Late reorders:** low-stock medicines are identified only by manual counting, and reorder requests are made verbally with no record of priority or status.
- **Scattered records:** supplier details, purchase invoices and batch details are kept in different registers.
- **No management view:** the pharmacy manager has no consolidated view of total stock, batch status and pending reorders.

1.3 Objectives

- To design a data model and forms for medicines, suppliers, purchases, batch inventory, stock issues, reorder requests and stock adjustments using Zoho Creator.
- To prevent entry of batches with an expiry date in the past and to automatically reduce a batch's available quantity when stock is issued, using Deluge scripting and form workflows.
- To send automatic e-mail notifications for low stock and for new reorder requests through Zoho Mail.
- To provide list reports (including an Urgent Reorders report) and a dashboard with key performance indicators and charts for stock and batch status.
- To configure role-based access through permission profiles, a role hierarchy and a sharing rule, and to document the deployment and governance requirements.

1.4 Scope of the Project

In scope: the Administrator and three working profiles — Pharmacy Admin, Inventory Staff and Pharmacist — and eight forms: Medicine, Create the Supplier form, Supplier, Purchase form, Batch Inventory form, Stock Issue, Reorder Request and Stock Adjustment. The application covers medicine and supplier master data, purchase recording, batch-wise inventory with expiry validation, stock issue with automatic quantity update, low-stock and reorder e-mail alerts, ten list reports and two dashboard pages.

Out of scope: point-of-sale billing and customer invoicing, barcode scanning, integration with accounting or ERP systems, a supplier portal, and automatic near-expiry scheduling and predictive demand forecasting. These are listed as future enhancements in Chapter 9.

1.5 Justification for a Low-Code Approach

A traditional implementation of this system would require a relational database for eight related entities, a backend with APIs, a web front end, a notification service, and user authentication with role-based access, which involves several weeks of work across several skills. Low-code application platforms provide these capabilities as configurable building blocks [1].

Zoho Creator suits this problem well. Forms create the data model with lookup relationships; reports and the Page builder provide list views, charts and KPI panels; Deluge expresses the expiry and stock-update rules; and form workflows with the send-mail task deliver notifications without a separate mail server [2]. Hosting, permission profiles, roles, sharing rules and a

mobile app are included, and pharmacy staff can adjust choices, thresholds and reports later without a development team [3].

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing system the medicine inventory process works as follows:

- Medicines received from a supplier are entered in a purchase register along with the invoice details.
- Stock is recorded in a stock register or spreadsheet, often maintained separately for each counter or store.
- Expiry dates are checked by physically inspecting strips and boxes, usually during periodic stock checks.
- Medicines issued for sale or internal use are noted on paper, or not recorded at all, so stock quantities are updated late.
- Low-stock medicines are identified by manual counting, and the supplier is contacted verbally or by phone.
- The manager collects information from several registers to understand the current stock position.

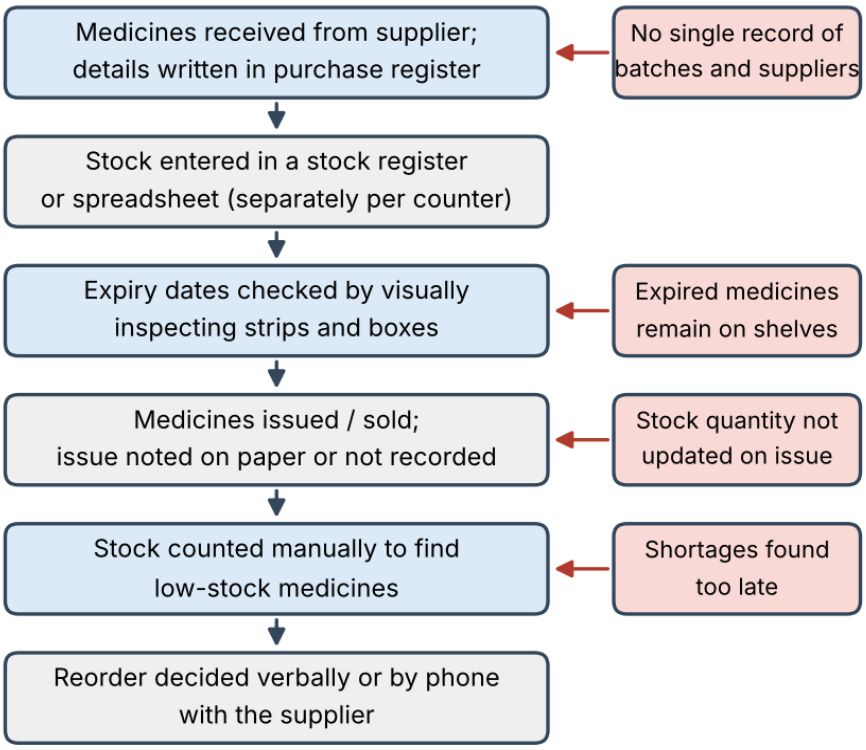


Figure 2.1: As-Is Process Flow of Manual Inventory and Expiry Tracking

2.2 Limitations of the Existing System

- Medicine, batch, purchase and supplier data is spread across registers and sheets, with no single source of truth.
- Expired batches are not detected automatically, and expired stock may be entered or remain available for issue.
- Available quantity is not reduced at the time of issue, which leads to mismatches between records and physical stock.
- There is no automatic alert for low stock, and reorder requests have no recorded priority, status or supplier.
- Stock write-offs for damaged, expired or returned medicines are not recorded with a reason and the person responsible.
- Management has no dashboard of total medicines, suppliers, stock and pending reorders.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Approximately 8–10 weeks (estimated) for eight entities, notification logic and web/mobile UI	Approximately 2–3 weeks: forms, reports and dashboard configured visually; Deluge for rules
Skills required	SQL, backend framework, front-end framework, mobile development, mail service and DevOps	Understanding of pharmacy inventory, Zoho Creator builders and basic Deluge
Hosting & infrastructure	Server or cloud set-up, database administration, backups and SSL	Hosted and maintained by Zoho
Mobile application	Separate mobile app or responsive UI must be built	Zoho Creator mobile app available from the same configuration
Security & access control	Authentication and role-based access coded by hand	Built-in permission profiles, roles and sharing rules
Maintenance & change	Code change and redeployment for each new field or status	Fields, choices, reports and workflows changed through builders
Customisation limits	Practically unlimited UI and logic customisation	Limited page and chart customisation; logic bound by Deluge and platform limits
Cost	Developer effort plus hosting and maintenance cost	Per-user subscription (free trial used for this project)

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	A. Sahay, A. Indamutsa, D. Di Ruscio and A. Pierantonio, 2020 [4]	Supporting the understanding and comparison of low-code development platforms	Taxonomy of low-code features (data modelling, business logic, workflow, deployment) and comparison of eight platforms.	Identified the platform capabilities the pharmacy application needs, especially data modelling and workflow.
2	R. Sanchis, Ó. García-Perales, F. Fraile and R. Poler, 2020 [5]	Low-code as enabler of digital transformation in manufacturing industry	Low-code helps operational teams digitise processes quickly and maintain them themselves.	Supports building a pharmacy stock application with low-code.
3	M. Hirzel, 2023 [6]	Low-code programming models (Communications of the ACM)	Classifies low-code programming models and describes their limits.	Guided the split between visual configuration and Deluge scripts (Section 8.5).
4	Central Drugs Standard Control Organisation, India [7]	Drugs and Cosmetics Act, 1940 and Rules, 1945	Sets requirements for storage, sale and supply of drugs, including that expired drugs must not be sold.	Justifies expiry validation and batch-wise records.
5	World Health Organization, 2011 [8]	Model guidance for the storage and transport of time- and temperature-sensitive pharmaceutical products (TRS 961, Annex 9)	Emphasises controlled storage, stock rotation and records for pharmaceutical products.	Supports storage location, storage instructions and batch status fields.

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
6	Gartner, Inc., 2024 [3]	Magic Quadrant for Enterprise Low-Code Application Platforms	Low-code platforms are mainstream for departmental business applications.	Justifies the low-code approach for a pharmacy management application.

2.5 Emerging Trends in Low-Code Development

Several low-code trends shaped this application [9]. Hyper-automation encouraged automating repetitive steps such as reducing stock on issue and sending alerts. Citizen development led to a design that pharmacy staff can maintain with short Deluge scripts and form workflows. Mobile access is available through the Zoho Creator mobile app. AI-assisted development with Zia was considered, and demand forecasting and scheduled near-expiry alerts are listed as future enhancements (Section 9.2).

2.6 Proposed System

The Smart Pharmacy Inventory & Expiry Alert System is a single Zoho Creator application with eight forms, ten list reports, two dashboard pages, four form workflows and role-based permission profiles. It addresses each limitation of Section 2.2 as follows:

- All medicine, supplier, purchase, batch, issue, reorder and adjustment data is stored once and linked through lookup fields.
- A validation workflow rejects batches whose expiry date is in the past, and the Batch Status field records Available, Reserved, Expired, Quarantined or Depleted batches.
- When stock is issued, the available quantity of the selected batch is reduced automatically.
- A low-stock alert e-mail is sent when the available quantity is 10 units or fewer, and a notification e-mail is sent when a reorder request is submitted.
- Reorder requests carry priority and status, and the Urgent Reorders report lists high-priority and urgent requests.
- Stock adjustments record the type, quantity, reason and the person who recorded them.
- A dashboard shows total medicines, total suppliers, total stock and pending reorders, with charts of stock by medicine and batch status split.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Pharmacy Admin	Pharmacy manager or administrator	Maintain medicines, suppliers and purchases; manage batch inventory, stock issues, reorder requests and stock adjustments; view the dashboard
Inventory Staff	Staff member who manages stock	Monitor and update batch inventory, record stock issues, submit reorder requests, view medicines and suppliers
Pharmacist	Staff member who dispenses medicines	View medicines and expiry details, record stock issues, view batch inventory shared by Inventory Staff, submit reorder requests

The Administrator (the application owner) manages users, permission profiles, roles, workflows and sharing rules and has full access to all components.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall maintain Medicine records with name, generic name, category, dosage form, strength, manufacturer, unit of measurement, reorder level, storage instructions and status.	Pharmacy Admin	High
FR-02	The system shall maintain Supplier records with type, contact person, phone, e-mail, address, city, GST number and status.	Pharmacy Admin	Medium
FR-03	The system shall record purchases with medicine, supplier, date, invoice number, quantity, unit cost, total cost and payment status.	Pharmacy Admin	High
FR-04	The system shall record batch inventory with batch number, manufacturing and expiry dates, quantity received, available quantity, unit cost, status, storage location and minimum stock level, linked to the medicine and purchase.	Pharmacy Admin, Inventory Staff	High
FR-05	The system shall reject a new batch whose expiry date is earlier than the current date.	System	High
FR-06	The system shall record stock issues (sale, internal use, damaged stock removal, other) against a batch with quantity, issued to and issued by.	Pharmacist, Inventory Staff	High
FR-07	The system shall automatically reduce the available quantity of the batch when stock is issued and sufficient stock exists.	System	High
FR-08	The system shall send a low-stock alert e-mail when the available quantity of the batch is 10 units or fewer.	System	High
FR-09	The system shall record reorder requests with current stock, reorder level, suggested quantity, supplier, priority and status, and notify by e-mail when a request is submitted.	Inventory Staff, Pharmacist	High
FR-10	The system shall provide an Urgent Reorders report for high-priority and urgent requests, sorted by priority and date.	Pharmacy Admin	Medium

FR ID	Requirement	Role(s)	Priority
FR-11	The system shall record stock adjustments (damaged, expired, returned to supplier, stock correction, customer return) with quantity, reason, recorded by and approval status.	Pharmacy Admin	Medium
FR-12	The system shall provide a list report for every form.	All profiles	Medium
FR-13	The system shall display a dashboard of total medicines, total suppliers, total stock and pending reorders, with charts of stock by medicine and batch status split.	Pharmacy Admin	High
FR-14	The system shall restrict each user to the forms and reports allowed by their permission profile and share batch inventory records from Inventory Staff to Pharmacists with read access.	All	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	The dashboard and list reports shall load within 3 seconds for up to 1,000 batch records.
NFR-02	Security	Users shall perform only the actions allowed by their permission profile; personal data fields such as supplier phone and e-mail shall be marked as personal data.
NFR-03	Usability	Forms shall use lookups and drop-downs to minimise typing, show medicine names in lookup lists, and be usable on the Zoho Creator mobile app.
NFR-04	Availability	The application shall be available 24 × 7 on Zoho's cloud, subject to Zoho's service availability.
NFR-05	Maintainability	Choices, thresholds, reports and workflows shall be changeable through the Zoho Creator builders.
NFR-06	Data integrity	Expired batches shall not be accepted at entry, and available quantity shall never be reduced below zero by the automatic stock update.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator, free trial edition
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
API testing tool	Not required – no external REST API is used in this version
Other Zoho services	Zoho Mail (e-mail notifications through the sendmail task)
Mobile	Zoho Creator mobile app (Android / iOS) – available; not separately configured
Time zone / date format	Asia/Kolkata; dd-MMM-yyyy; 24-hour time
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

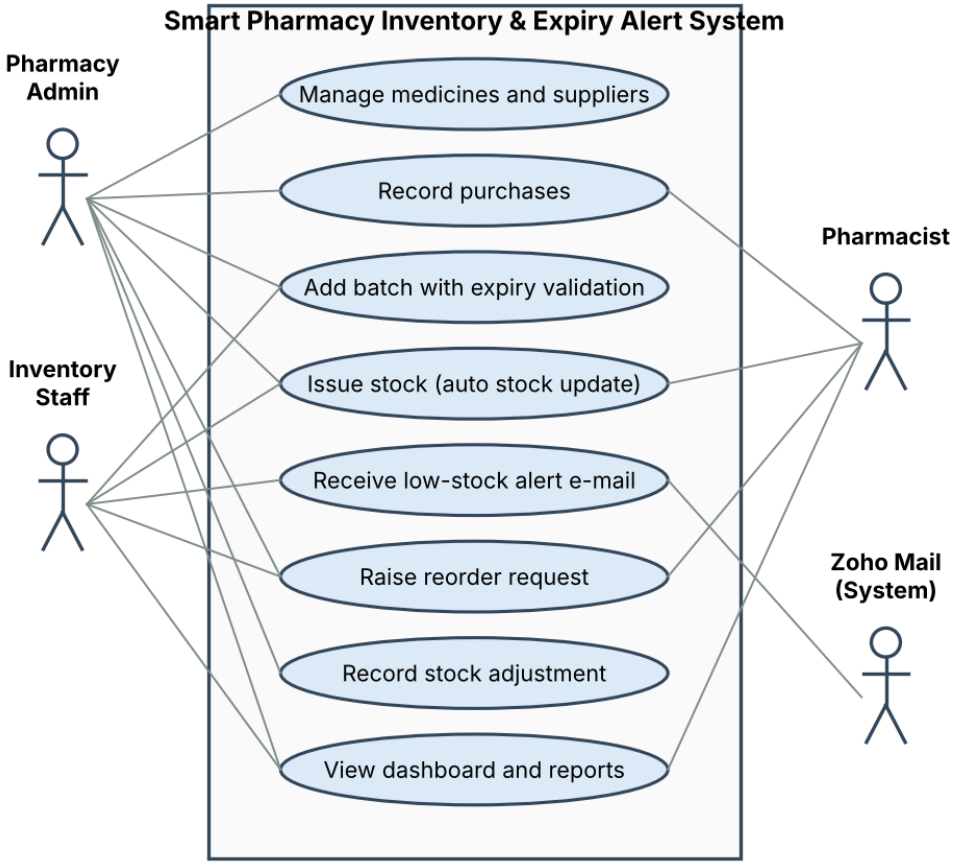


Figure 3.1: Use Case Diagram of the Smart Pharmacy Inventory & Expiry Alert System

Table 3.5: Use Case Description – Issue Stock and Trigger Alerts

Use Case ID	UC-01
Use Case Name	Issue Stock and Trigger Alerts
Actor(s)	Pharmacist, Inventory Staff (primary); System
Pre-condition	The medicine and a batch with available quantity exist in Batch Inventory.
Main Flow	1. The user opens the Stock Issue form and selects the batch. 2. The user enters issue date, quantity issued, issue type, issued to and issued by, and submits. 3. DS-02 reduces the batch's available quantity by the quantity issued. 4. If the batch's available quantity is 10 or fewer, DS-03 sends the Low Stock Alert e-mail. 5. The Inventory Staff raises a Reorder Request for the medicine.
Alternate Flow	3a. The quantity issued is greater than the available quantity: the quantity is not reduced and the message "Insufficient stock available." is shown.
Post-condition	The batch's available quantity reflects the issue, and a low-stock e-mail has been sent if the threshold was reached.
Related FR	FR-06, FR-07, FR-08

Table 3.6: Use Case Description – Raise and Process a Reorder Request

Use Case ID	UC-02
Use Case Name	Raise and Process a Reorder Request
Actor(s)	Inventory Staff, Pharmacist, Pharmacy Admin; System
Pre-condition	A medicine is at or below its reorder level.
Main Flow	1. The user opens Reorder Request, selects the medicine and enters current stock, reorder level, suggested order quantity, supplier, priority and remarks. 2. On submission, DS-04 sends the Medicine Reorder Request e-mail. 3. The Pharmacy Admin reviews the request in All Requests or Urgent Reorders. 4. The request status is updated through Pending → Approved → Ordered → Received.
Alternate Flow	4a. The request is set to Rejected with remarks.
Post-condition	The request is tracked with priority and status until it is Received or Rejected.
Related FR	FR-09, FR-10

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

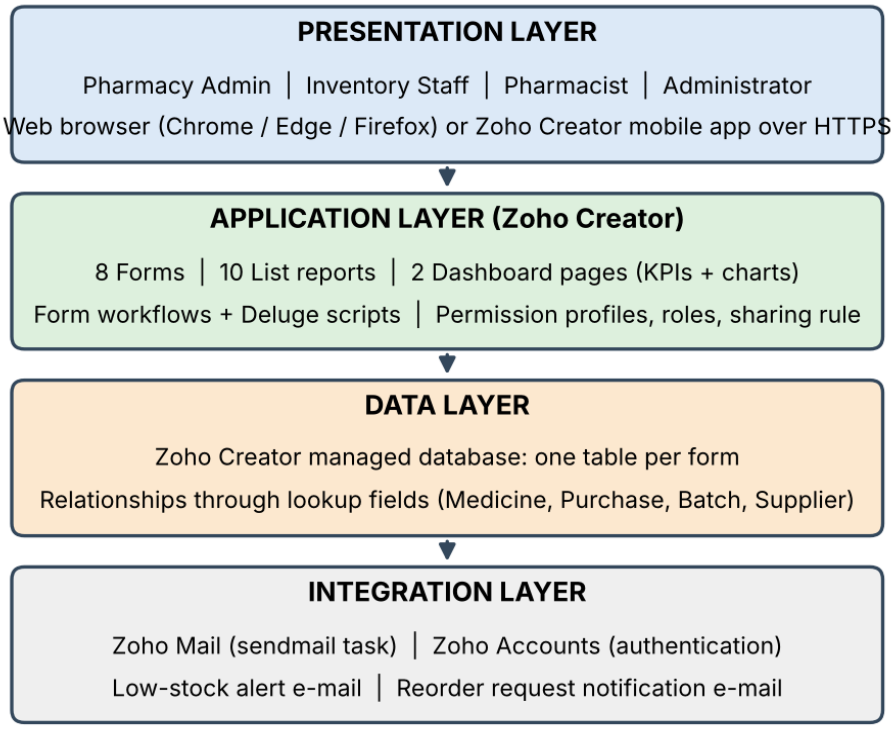


Figure 4.1: System Architecture of the Smart Pharmacy Inventory & Expiry Alert System

Figure 4.1 shows the four-layer architecture. In the presentation layer, the Pharmacy Admin, Inventory Staff, Pharmacist and the Administrator use the application through a web browser or the Zoho Creator mobile app over HTTPS. The application layer is the Zoho Creator application: eight forms capture data, list reports and two dashboard pages present it, Deluge scripts and form workflows implement the expiry and stock-update rules, and permission profiles, roles and a sharing rule control access.

The data layer is Zoho Creator's managed database, where each form is a table and relationships are held through lookup fields. The integration layer contains Zoho Mail, which delivers the low-stock and reorder notification e-mails, and Zoho Accounts for authentication.

4.2 Data Model

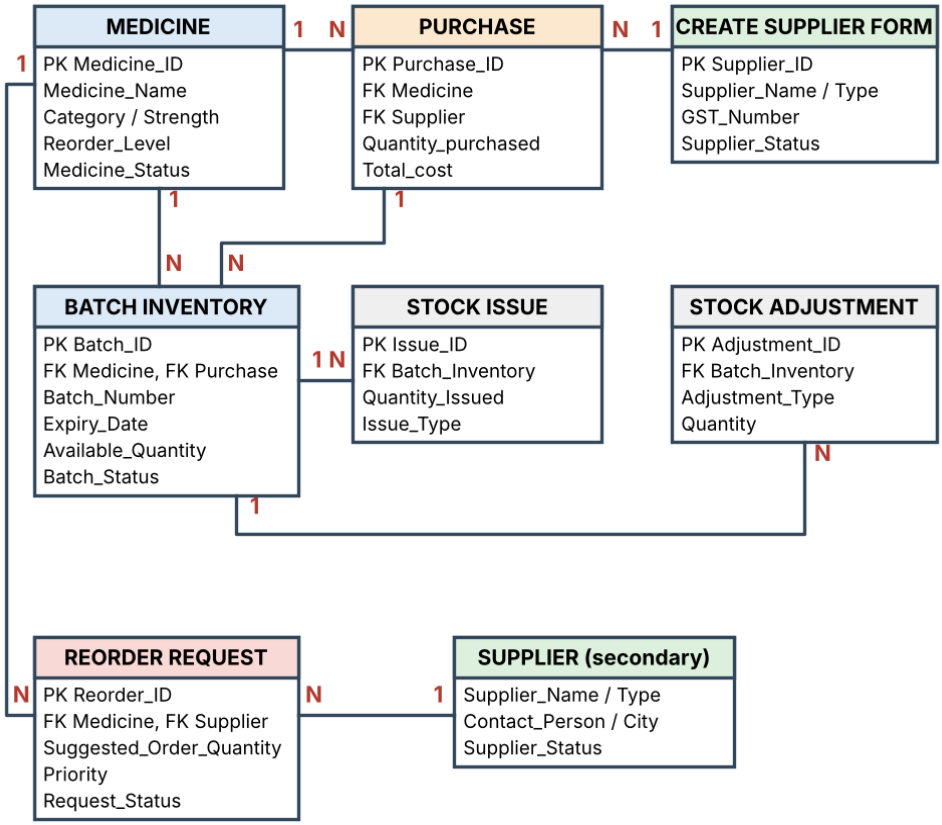


Figure 4.2: Entity–Relationship Diagram of the Smart Pharmacy Inventory & Expiry Alert System

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Medicine	Medicine_ID, Medicine_Name, Reorder_Level	Purchase form	One-to-Many	Lookup (Medicine) in Purchase form
Medicine	Medicine_ID	Batch Inventory form	One-to-Many	Lookup (Medicine) in Batch Inventory form
Medicine	Medicine_ID	Reorder Request	One-to-Many	Lookup (Medicine) in Reorder Request
Create the Supplier form	Supplier_ID, Supplier_Name, GST_Number	Purchase form	One-to-Many	Lookup (Supplier name) in Purchase form
Supplier	Supplier_Name, Supplier_Type	Reorder Request	One-to-Many	Lookup (Supplier) in Reorder Request
Purchase form	Purchase_ID, Invoice_Number	Batch Inventory form	One-to-Many	Lookup (Purchase form) in Batch Inventory form

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Batch Inventory form	Batch_ID, Expiry_Date, Available_Quantity	Stock Issue	One-to-Many	Lookup (Batch Inventory form) in Stock Issue
Batch Inventory form	Batch_ID	Stock Adjustment	One-to-Many	Lookup (Batch Inventory form) in Stock Adjustment

Medicine and Batch Inventory are the central entities. Every batch belongs to one medicine and can be traced to the purchase that brought it in; every stock issue and stock adjustment refers to one batch; and reorder requests refer to a medicine and a supplier. The application contains two supplier forms: Create the Supplier form (with GST number and structured address) is used by the Purchase form, and the Supplier form is used by Reorder Request. Consolidating them is listed in the defect and improvement log (Table 6.2).

4.3 Form Design

Tables 4.2 to 4.9 list the fields of the eight forms using Zoho Creator field types, including Auto Number, Single Line, Multi Line, Number, Date, Currency (INR), Drop Down, Lookup, Email, Phone and Address. The export of the application does not mark individual fields as mandatory, so the tables list the field type and the choices or validation that are configured.

Table 4.2: Field Specification – Medicine Form

Field Label	Link Name	Field Type	Choices / Validation / Remarks
Medicine ID	Medicine_ID	Auto Number	System generated
Medicine Name	Medicine_Name	Single Line	Name of the medicine; shown in lookup lists
Generic Name	Generic_Name	Single Line	Generic (salt) name
Category	Category	Drop Down	Tablet / Capsule / Syrup / Injection / Ointment / Other
Dosage Form	Dosage_Form	Drop Down	Tablet / Capsule / Liquid / Injectable / Topical / Other
Strength	Strength	Single Line	e.g., 500 mg
Manufacturer	Manufacturer	Single Line	Manufacturer details
Unit of Measurement	Unit_of_Measurement	Drop Down	Strip / Bottle / Box / Vial / Tube / Piece
Reorder Level	Reorder_Level	Number	Stock level at which reorder is needed
Storage Instructions	Storage_Instructions	Multi Line	Storage conditions
Medicine Status	Medicine_Status	Drop Down	Active / Inactive

Table 4.3: Field Specification – Create the Supplier Form

Field Label	Link Name	Field Type	Choices / Validation / Remarks
Supplier ID	Supplier_ID	Auto Number	System generated
Supplier Name	Supplier_Name	Single Line	Shown in the Purchase form lookup
Supplier Type	Supplier_Type	Drop Down	Manufacturer / Wholesaler / Distributor / Retail Supplier
Contact Person	Contact_Person	Single Line	Supplier contact

Field Label	Link Name	Field Type	Choices / Validation / Remarks
Phone Number	Phone_Number1	Phone	Marked as personal data
Email Address	Email_Address	Email	Maximum 80 characters; marked as personal data
Address	Address	Address	Address lines, city/district, state, postal code, country; coordinates captured
City	City	Single Line	Supplier city
GST Number	GST_Number	Single Line	Supplier GST registration number
Supplier Status	Supplier_Status	Drop Down	Active / Inactive / Pending Verification

Table 4.4: Field Specification – Supplier Form

Field Label	Link Name	Field Type	Choices / Validation / Remarks
Supplier Name	Supplier_Name	Single Line	Supplier name; shown in the Reorder Request lookup
Supplier Type	Supplier_Type	Drop Down	Manufacturer / Wholesaler / Distributor / Retail Supplier
Contact Person	Contact_Person	Single Line	Supplier contact
Phone Number	Phone_Number1	Phone	Marked as personal data
Email Address	Email_Address	Email	Marked as personal data
Address	Address	Multi Line	Supplier address
City	City	Single Line	Supplier city
Supplier Status	Supplier_Status	Drop Down	Active / Inactive / Pending Verification

Table 4.5: Field Specification – Purchase Form

Field Label	Link Name	Field Type	Choices / Validation / Remarks
Purchase ID	Purchase_ID	Auto Number	System generated
Medicine	Medicine	Lookup (Medicine)	Displays Medicine Name; sorted ascending
Supplier	Create_the_Supplier_form	Lookup (Create the Supplier form)	Displays Supplier Name; sorted ascending
Purchase Date	Purchase_Date	Date	Format dd-MMM-yyyy
Invoice Number	Invoice_Number	Single Line	Supplier invoice reference
Quantity purchased	Quantity_purchased	Number	Quantity bought
Unit cost	Unit_cose	Currency (INR)	Indian comma format
Total cost	Total_cost	Currency (INR)	Indian comma format
Payment status	Payment_status	Single Line	Payment state (e.g., Paid / Pending)

Table 4.6: Field Specification – Batch Inventory Form

Field Label	Link Name	Field Type	Choices / Validation / Remarks
Batch ID	Batch_ID	Auto Number	System generated
Medicine	Medicine	Lookup (Medicine)	Displays Medicine Name
Purchase form	Purchase_form	Lookup (Purchase form)	Displays Purchase ID

Field Label	Link Name	Field Type	Choices / Validation / Remarks
Batch Number	Batch_Number	Single Line	Manufacturer batch number
Manufacturing Date	Manufacturing_Date	Date	Format dd-MMM-yyyy
Expiry Date	Expiry_Date	Date	Validated by DS-01: cannot be in the past
Quantity Received	Quantity_Received	Number	Quantity received in the batch
Available Quantity	Available_Quantity	Number	Reduced by DS-02 on stock issue
Unit Cost	Unit_Cost	Currency (INR)	Indian comma format
Batch Status	Batch_Status	Drop Down	Available / Reserved / Expired / Quarantined / Depleted
Storage Location	Storage_Location	Single Line	Shelf / rack / store location
Minimum Stock Level	Minimum_Stock_Level	Number	Minimum quantity to keep

Table 4.7: Field Specification – Stock Issue Form

Field Label	Link Name	Field Type	Choices / Validation / Remarks
Issue ID	Issue_ID	Auto Number	System generated
Issue Date	Issue_Date	Date	Format dd-MMM-yyyy
Batch Inventory form	Batch_Inventory_form	Lookup (Batch Inventory form)	Batch from which stock is issued
Quantity Issued	Quantity_Issued	Number	Used by DS-02
Issued To	Issued_To	Single Line	Recipient (customer / department)
Issue Type	Issue_Type	Drop Down	Sale / Internal Use / Damaged Stock Removal / Other
Issued By	Issued_By	Single Line	Staff member issuing stock
Remarks	Remarks	Multi Line	Additional information

Table 4.8: Field Specification – Reorder Request Form

Field Label	Link Name	Field Type	Choices / Validation / Remarks
Reorder ID	Reorder_ID	Auto Number	System generated
Request Date	Request_Date	Date	Format dd-MMM-yyyy
Medicine	Medicine	Lookup (Medicine)	Medicine to reorder
Current Stock	Current_Stock	Number	Stock at the time of request
Reorder Level	Reorder_Level	Number	Reorder level of the medicine
Suggested Order Quantity	Suggested_Order_Quantity	Number	Quantity to order
Supplier	Supplier	Lookup (Supplier)	Supplier to order from
Priority	Priority	Drop Down	Low / Medium / High / Urgent
Request Status	Request_Status	Drop Down	Pending / Approved / Ordered / Received / Rejected
Remarks	Remarks	Multi Line	Additional information

Table 4.9: Field Specification – Stock Adjustment Form

Field Label	Link Name	Field Type	Choices / Validation / Remarks
Adjustment ID	Adjustment_ID	Auto Number	System generated
Adjustment Date	Adjustment_Date	Date	Format dd-MMM-yyyy
Batch Inventory form	Batch_Inventory_form	Lookup (Batch Inventory form)	Batch being adjusted
Adjustment Type	Adjustment_Type	Drop Down	Damaged / Expired / Returned to Supplier / Stock Correction / Customer Return
Quantity	Quantity	Number	Quantity adjusted
Reason	Reason	Multi Line	Reason for adjustment
Recorded By	Recorded_By	Single Line	Staff member recording the adjustment
Approval Status	Approval_Status1	Drop Down	Pending / Approved / Rejected
Remarks	Remarks	Multi Line	Additional information
Approval Status (extra) / Drop Down	Approval_Status, Dropdown1	Drop Down	Placeholder choices (Choice 1–3) left from form design; to be removed (BUG-01)

4.4 Process and Workflow Design

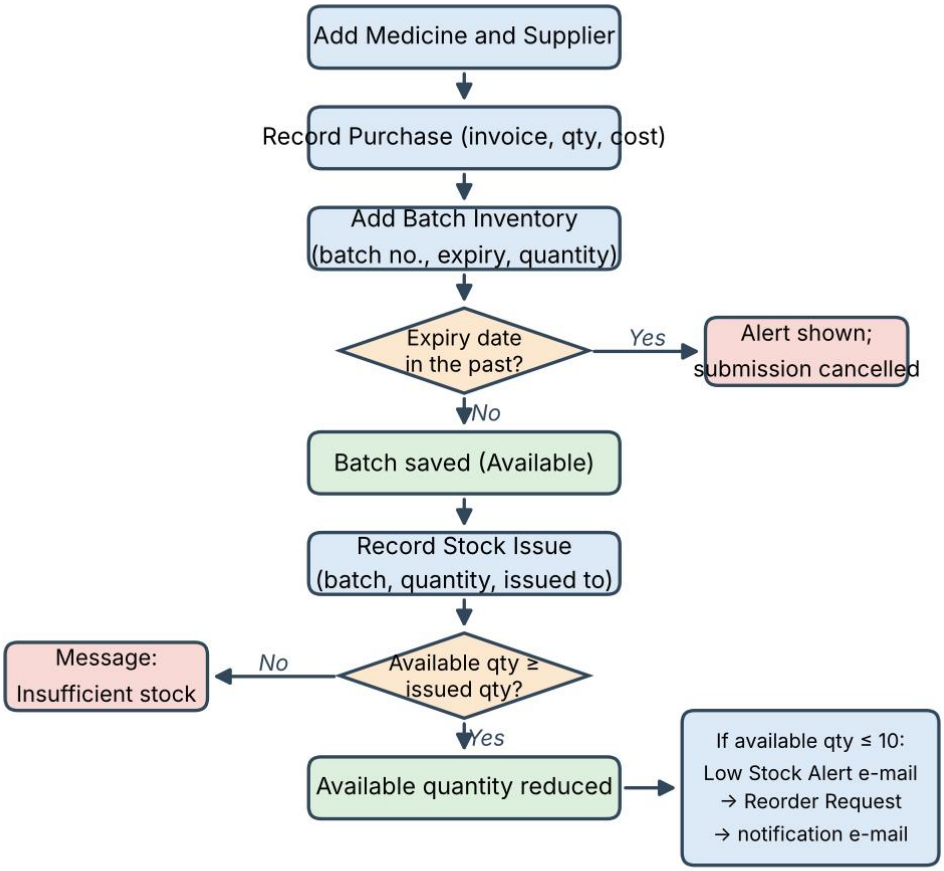


Figure 4.3: Stock Entry, Issue and Alert Process Flow

Figure 4.3 illustrates the inventory process. Medicines and suppliers are created as master data, a purchase is recorded, and the received stock is added as a batch. On submission of the batch, the expiry check rejects any batch whose expiry date is in the past. When stock is issued, the available quantity of the batch is reduced if sufficient stock exists; otherwise the user is told that stock is insufficient. When the available quantity is 10 units or fewer, the Low Stock Alert e-mail is sent. The Inventory Staff then raises a reorder request, which triggers a notification e-mail and is tracked through Pending, Approved, Ordered and Received. Damaged, expired or returned stock is recorded in the Stock Adjustment form.

4.5 User Interface and Navigation Design

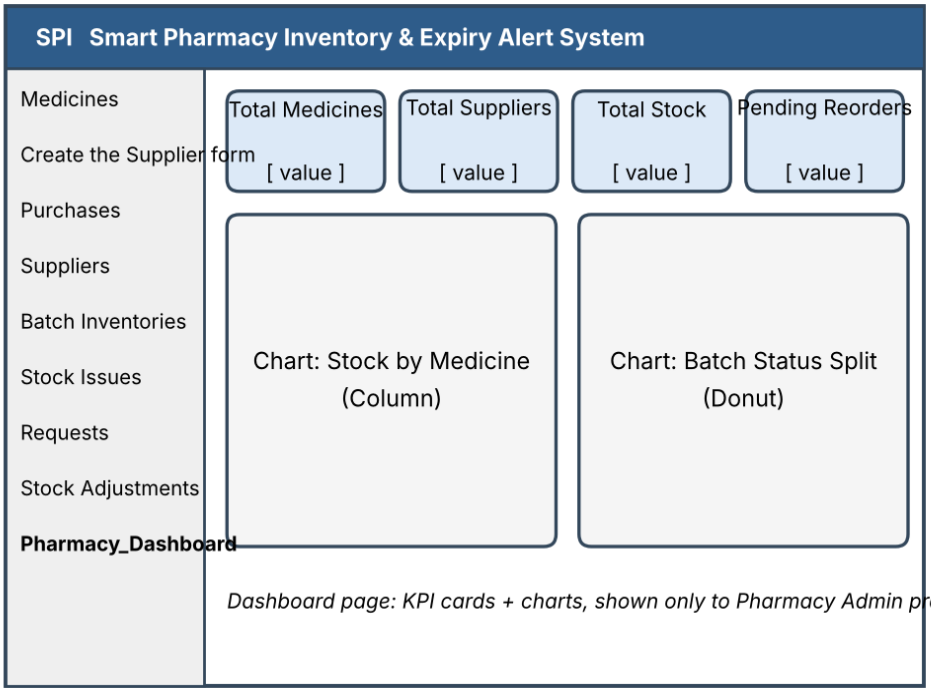


Figure 4.4: Navigation Structure and Wireframe of the Dashboard

The navigation menu groups the application into the sections Medicines, Create the Supplier form, Purchases, Suppliers, Batch Inventories, Stock Issues, Requests, Stock Adjustments and Pharmacy_Dashboard, together with the standard Approvals and Reports sections provided by Zoho Creator. Each section contains its form and the corresponding reports. The dashboard pages show KPI panels for Total Medicines, Total Suppliers, Total Stock and Pending Reorders, a column chart of stock by medicine and a donut chart of the batch status split.

4.6 Access Control Design

Table 4.10: Role–Permission Matrix

Component	Pharmacist	Inventory Staff	Pharmacy Admin
Medicine	C V	C V	C V E

Component	Pharmacist	Inventory Staff	Pharmacy Admin
Create the Supplier form	—	—	C V E
Supplier	C V	C V	C V E
Purchase form	—	—	C V E
Batch Inventory form	C V (shared)	C V E	C V E
Stock Issue	C V E	C V E	C V E
Reorder Request	C V E	C V E	C V E
Stock Adjustment	—	—	C V E
Pharmacy_Dashboard	—	—	V

C = Create, V = View, E = Edit. The Administrator has full access to all components, workflows and settings. Roles are arranged as CEO → Pharmacist → Inventory Staff, with Pharmacy Admin also reporting to CEO; a sharing rule shares Batch Inventory records from Inventory Staff to the Pharmacist role with read access.

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application was created in a Zoho Creator workspace using the low-code development approach and named "Smart Pharmacy Inventory & Expiry Alert System" (application icon text "SPI"). It was developed from scratch by creating the required forms, reports, workflows and dashboard pages. The application was designed to manage medicines, suppliers, purchases, batch inventory with expiry dates, stock issues, reorder requests and stock adjustments.

The application menu is organised into sections: Medicines, Create the Supplier form, Purchases, Suppliers, Batch Inventories, Stock Issues, Requests, Stock Adjustments and Pharmacy_Dashboard. The forms are connected using lookup fields wherever related records are required, and sample medicine, supplier, purchase and batch records were added for testing. The application uses the Asia/Kolkata time zone, the dd-MMM-yyyy date format and 24-hour time.

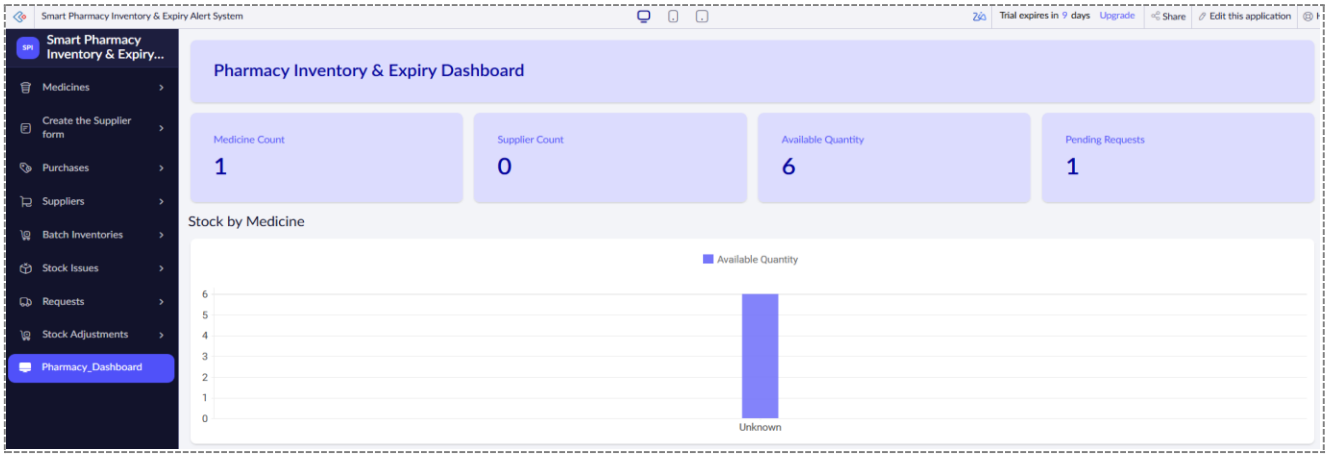


Figure 5.1: Smart Pharmacy Inventory & Expiry Alert System Application Home and Dashboard

5.2 Forms and Data Capture

The eight forms were created in Zoho Creator to collect and maintain pharmacy information: Medicine, Create the Supplier form, Supplier, Purchase form, Batch Inventory form, Stock Issue, Reorder Request and Stock Adjustment. Their field details are presented in Tables 4.2–4.9. The following features were configured:

- **Field types:** Auto Number, Single Line, Multi Line, Number, Date, Currency (INR with Indian comma format), Drop Down, Lookup, Email, Phone and Address fields were used according to the data being captured — eleven distinct field types.

- **Lookup relationships:** Purchase links to Medicine and to the supplier; Batch Inventory links to Medicine and Purchase; Stock Issue and Stock Adjustment link to Batch Inventory; Reorder Request links to Medicine and Supplier. Lookup lists display readable values such as Medicine Name and Supplier Name, sorted in ascending order.
- **Drop-down choices:** Predefined choices for category, dosage form, unit of measurement, batch status, issue type, adjustment type, priority and request status keep data consistent.
- **Auto numbering:** Medicine ID, Supplier ID, Purchase ID, Batch ID, Issue ID, Reorder ID and Adjustment ID are generated automatically.
- **Personal data protection:** Supplier phone number and e-mail address fields are marked as personal data.
- **Success messages:** Forms such as Batch Inventory and Purchase display confirmation messages after submission.
- **Data submission:** Each form provides Submit on add and Update/Cancel on edit.

No subform, image or file upload field was required in this version of the application.

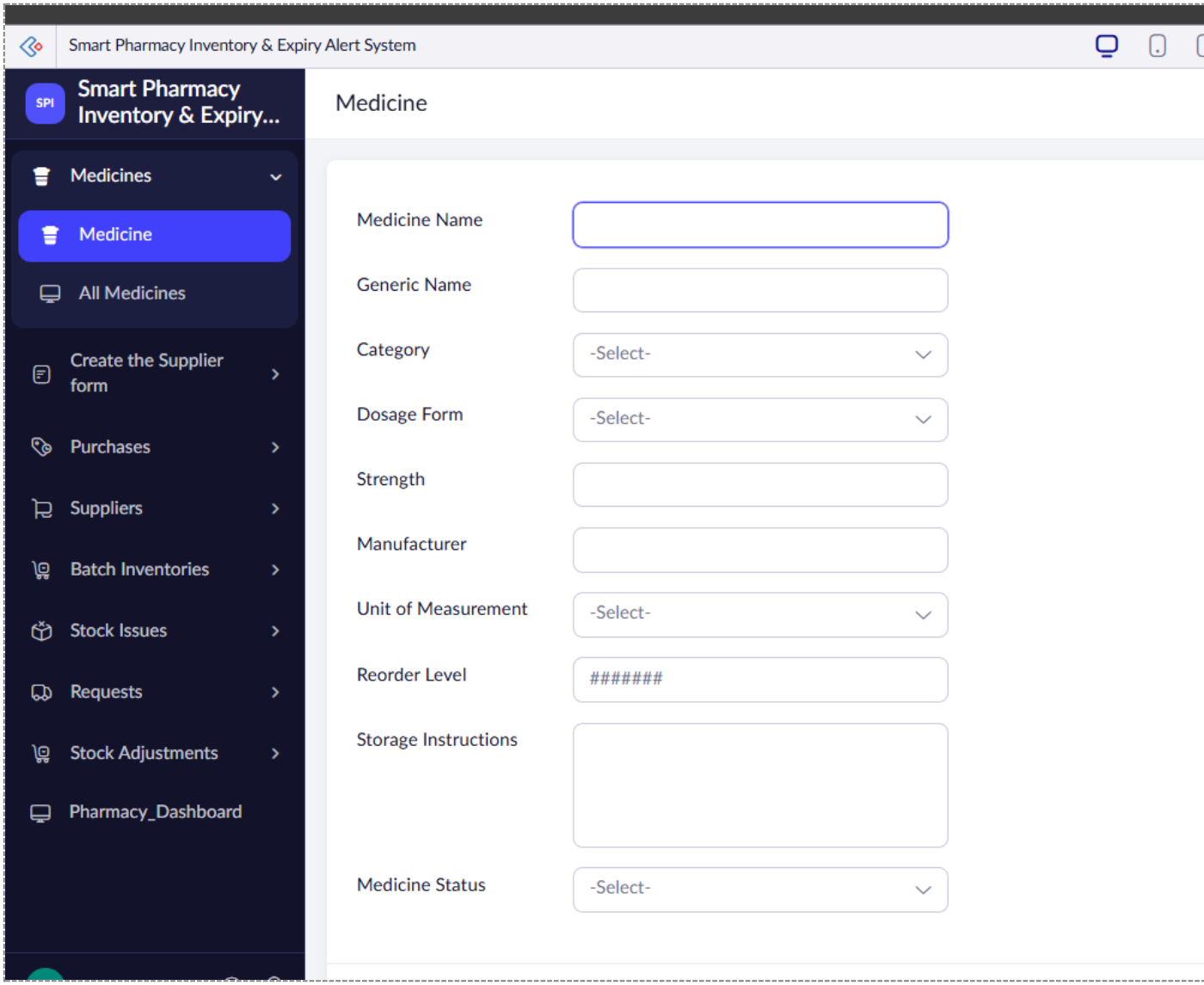


Figure 5.2: Medicine Form in the Form Builder

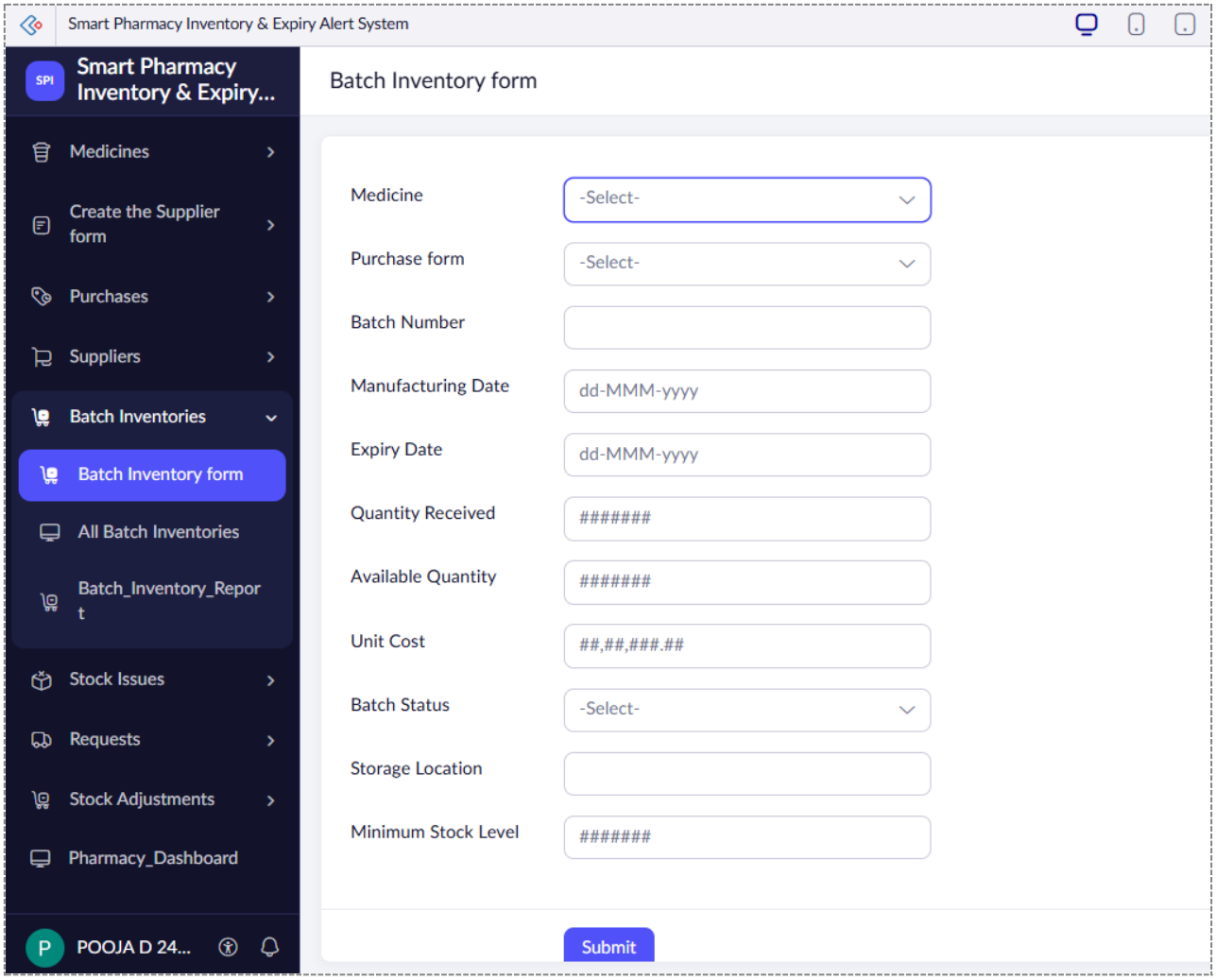


Figure 5.4: Batch Inventory Form in the Form Builder

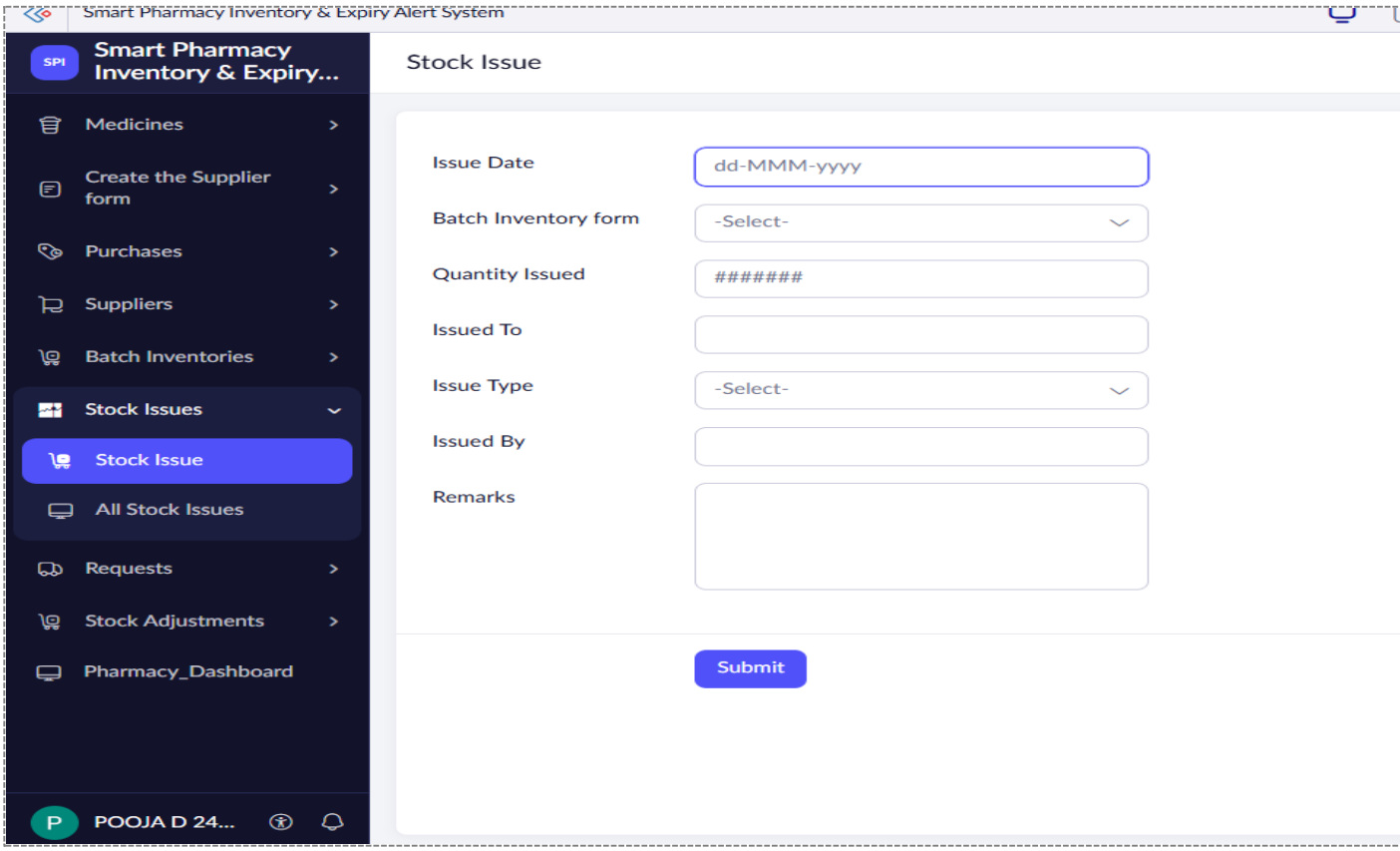


Figure 5.6: Stock Issue Form in the Form Builder

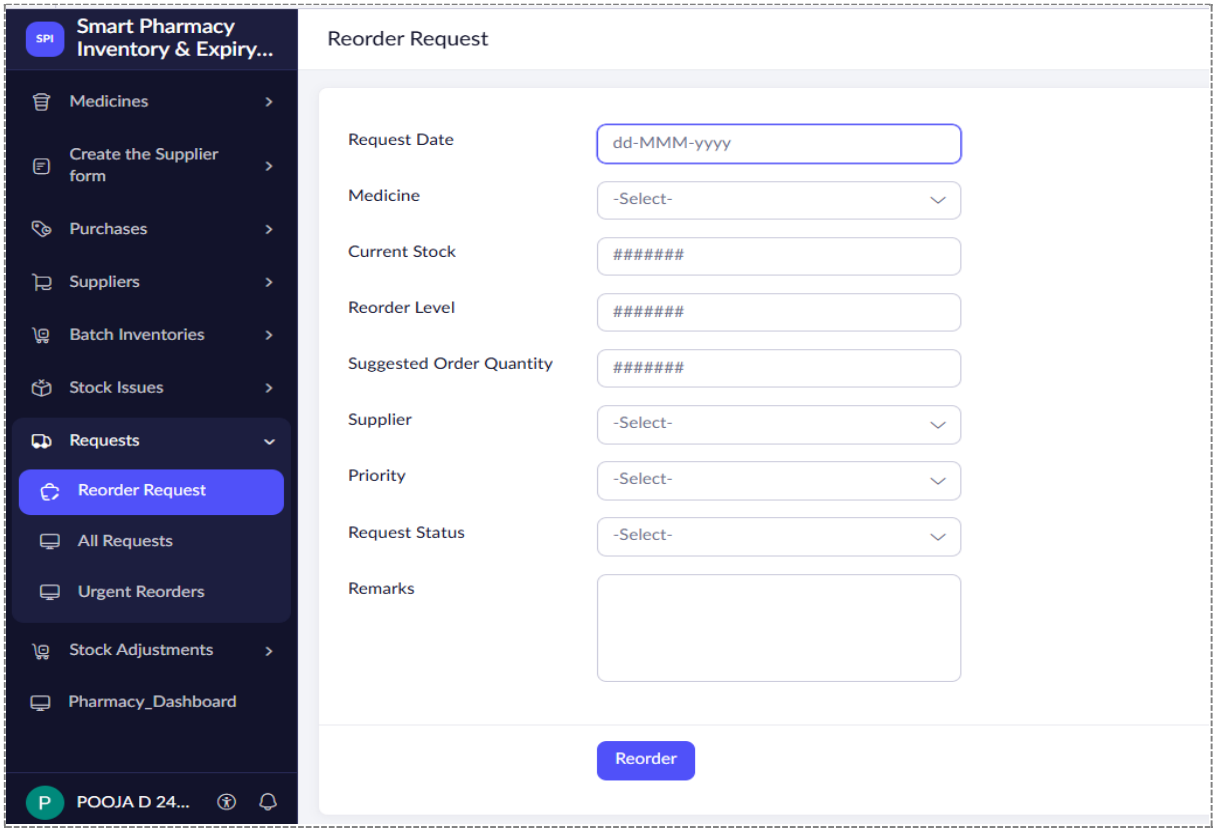


Figure 5.8: Reorder Request Form in the Form Builder

5.3 Reports

Each form has a list report, and two additional custom list reports were created — Batch_Inventory_Report and Urgent Reorders (Table 5.1). The Urgent Reorders report filters reorder requests by status and priority and sorts them by priority and request date in descending order, so that the most critical requests appear first. Phone numbers in the supplier report are shown in plain-number format and addresses can be shown as text or linked to a map.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All Medicines	List	Medicine	Default list (11 columns)	—	All
Create the Supplier form Report	List	Create the Supplier form	Default list; address as text/map link	—	Admin
All Purchases	List	Purchase form	Default list	—	Admin
All Suppliers	List	Supplier	Default list	—	All
All Batch Inventories	List	Batch Inventory form	Default list	—	All
Batch_Inventory_Report	List	Batch Inventory form	All rows, all batch columns	—	—
All Stock Issues	List	Stock Issue	Default list	—	All
All Requests	List	Reorder Request	Default list	—	All
All Stock Adjustments	List	Stock Adjustment	Default list	—	Admin
Urgent Reorders	List	Reorder Request	Filter: Urgent Pending; sort: Priority desc, Request Date desc	—	—

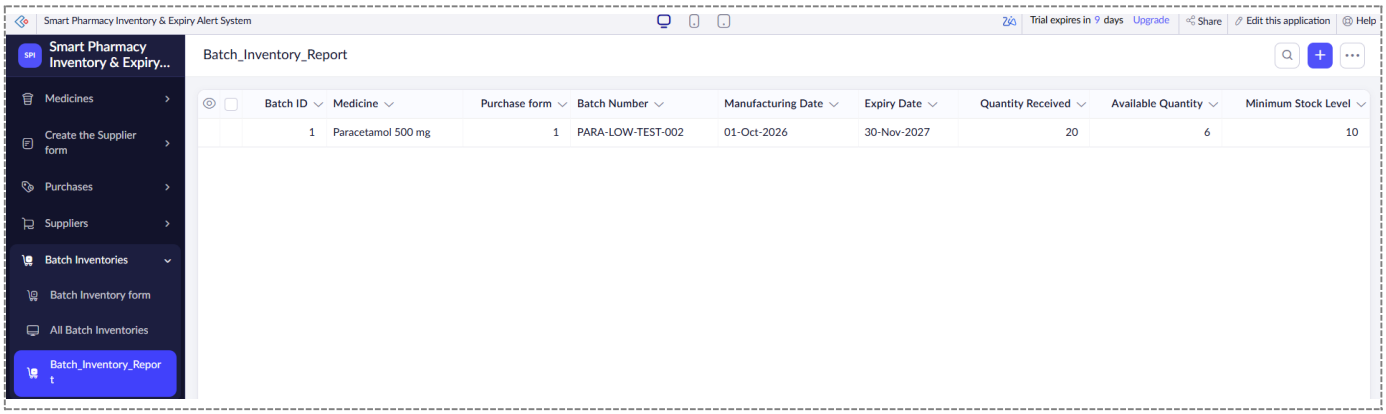


Figure 5.10: All Batch Inventories Report

5.4 Pages and Dashboards

Two dashboard pages were built with the Page builder:

- **Pharmacy_Dashboard:** KPI panels showing the count of medicines, the count of suppliers, total stock (sum of available quantity of all batches) and pending reorders, with a bar chart of available quantity by batch number.
- **Pharmacy Dashboard 2:** a title bar, four KPI cards — Total Medicines, Total Suppliers, Total Stock and Pending Reorders (reorder requests whose status is Pending)

— and two charts: Stock by Medicine (column chart of available quantity) and Batch Status Split (donut chart of the number of batches in each batch status).

The KPI values are calculated from live data in the application, so they change as medicines, batches, stock issues and reorder requests are added. The dashboard is made visible to the Pharmacy Admin profile.

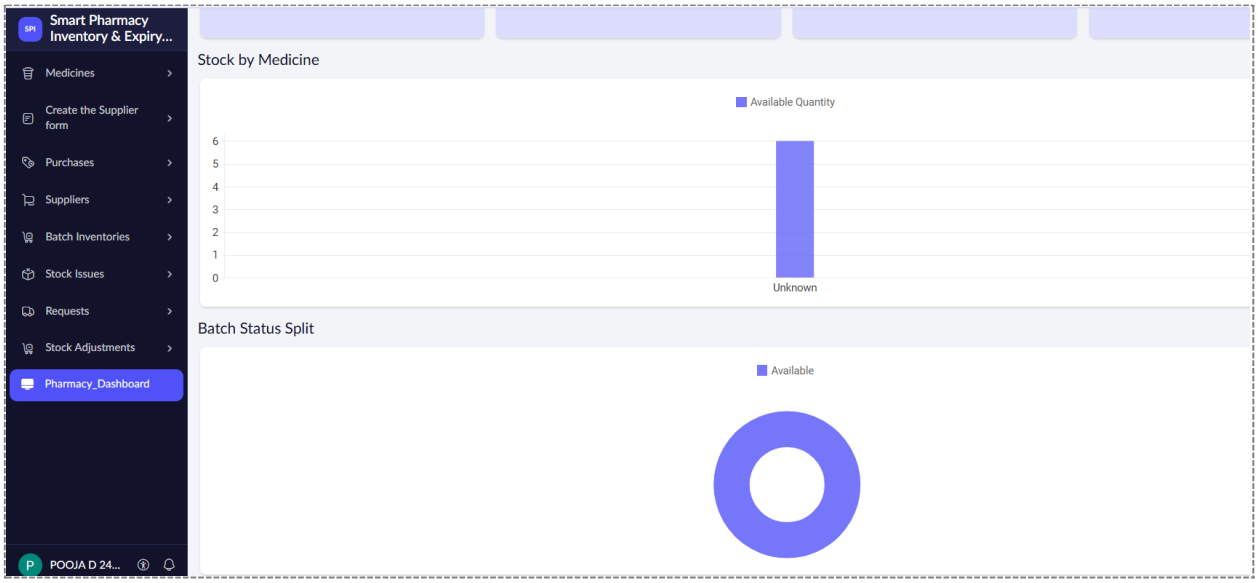


Figure 5.11: Pharmacy Dashboard 2 with KPI Cards and Charts

5.5 Deluge Scripting

Deluge scripting is used to automate the expiry check and the stock update, while send-mail actions deliver notifications. The scripts work with form events, record fetches, date comparison and record updates. Table 5.2 presents the scripts and automation actions used in the application.

Table 5.2: Deluge Scripts and Automation Actions Implemented

ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Batch Inventory form → On add → On validate	Reject a batch whose expiry date is in the past	null check, date comparison with zoho.currentdate, alert, cancel submit
DS-02	Stock Issue → On add → On success	Reduce available quantity of the batch on issue	fetch record by ID, count(), if-else, arithmetic, field update, info
DS-03	Stock Issue → On add or edit → On success (criteria: Available_Quantity ≤ 10)	Send Low Stock Alert e-mail	workflow criteria, sendmail task, zoho.adminuserid
DS-04	Reorder Request → On add → On success	Send Medicine Reorder Request e-mail	sendmail task, zoho.adminuserid

Code 5.1: Expiry date validation (DS-01)

```
// DS-01 | Form: Batch_Inventory_form | Event: On add - On validate
// Purpose: Do not accept a batch whose expiry date is in the past
if(input.Expiry_Date != null && input.Expiry_Date < zoho.currentdate)
{
    alert "Expiry date cannot be in the past.";
    cancel submit;
}
```

Code 5.2: Automatic stock update (DS-02)

```
// DS-02 | Form: Stock_Issue | Event: On add - On success
// Purpose: Reduce the available quantity of the issued batch
if(input.Batch_Inventory_form != null && input.Quantity_Issued > 0)
{
    batchRecord = Batch_Inventory_form[ID == input.Batch_Inventory_form];
    if(batchRecord.count() > 0)
    {
        currentQty = batchRecord.Available_Quantity;
        issuedQty = input.Quantity_Issued;
        if(currentQty >= issuedQty)
        {
            batchRecord.Available_Quantity = currentQty - issuedQty;
            info "Stock quantity updated.";
        }
        else
        {
            info "Insufficient stock available.";
        }
    }
}
```

Code 5.3: Low stock alert e-mail (DS-03)

```
// DS-03 | Form: Stock_Issue | Event: On add or edit - On success
// Criteria: Batch_Inventory_form.Available_Quantity <= 10
sendmail
[
    from : zoho.adminuserid
    to : "pooja.d.2024.cse@rajalakshmi.edu.in"
    subject : "Low Stock Alert – Pharmacy Inventory"
    message : "Warning: Available medicine stock has reached 10 or fewer units.
    Please arrange restocking."
]
```

DS-01: the script checks that an expiry date has been entered and compares it with the current date. If the date has already passed, an alert is shown and the submission is cancelled, so an expired batch cannot enter the inventory.

DS-02: after a stock issue is saved, the script fetches the selected batch, compares the available quantity with the quantity issued and, if enough stock exists, subtracts the issued quantity from the batch. If stock is insufficient, the quantity is not reduced and a message is shown.

DS-03 and DS-04: the send-mail tasks use the administrator's account as the sender, so no password or credential is stored in the application.

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Batch_Inventory_Expiry_Check	Form Workflow	Batch Inventory form: on add, on validate	Expiry date earlier than today	Alert; cancel submit	DS-01
Automatic Stock Update	Form Workflow	Stock Issue: on add, on success	Batch selected and quantity issued > 0	Reduce available quantity	DS-02
Low Stock Alert	Form Workflow	Stock Issue: on add or edit, on success	Available_Quantity ≤ 10	Send Low Stock Alert e-mail	DS-03
Reorder Request Notification	Form Workflow	Reorder Request: on add, on success	—	Send Medicine Reorder Request e-mail	DS-04

Approvals, schedules and Blueprints. No approval process, scheduled workflow or Blueprint was configured in this version. Approval of stock adjustments and reorder requests is currently handled through the Approval Status and Request Status fields, and scheduled near-expiry alerts and approval workflows are listed as future enhancements (Section 9.2). Batch workflows were not required.

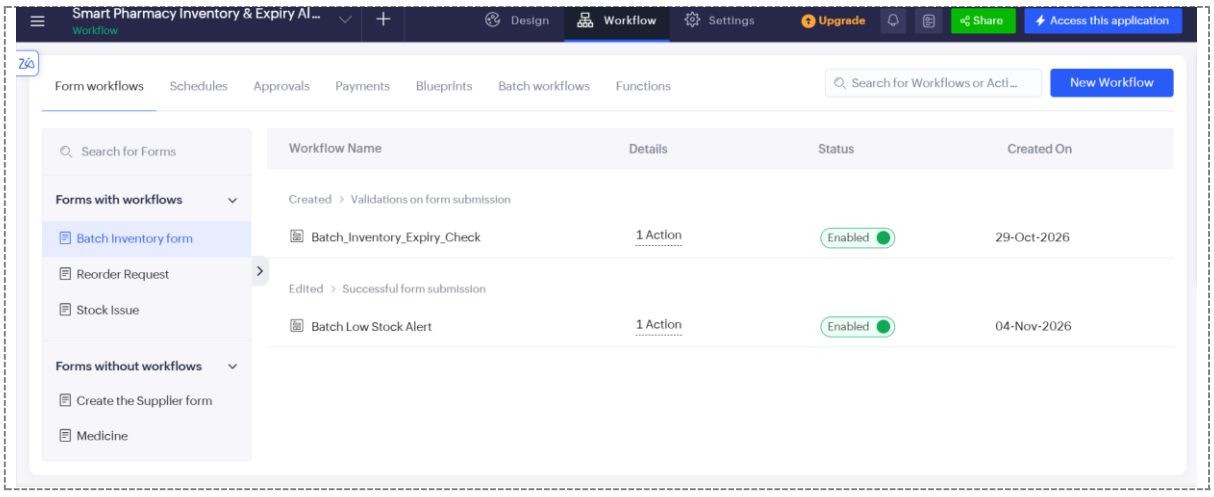


Figure 5.12: Workflow List of the Application

5.7 Integrations

The application is integrated with Zoho Mail through the built-in send-mail task. Two notifications are configured. The Low Stock Alert e-mail with the subject “Low Stock Alert – Pharmacy Inventory” is sent when a stock issue leaves the batch with 10 or fewer units, and the Medicine Reorder Request e-mail is sent when a new reorder request is submitted. Both are sent from the administrator’s account to the configured pharmacy recipient. No external REST API, connection or webhook was required in this version.

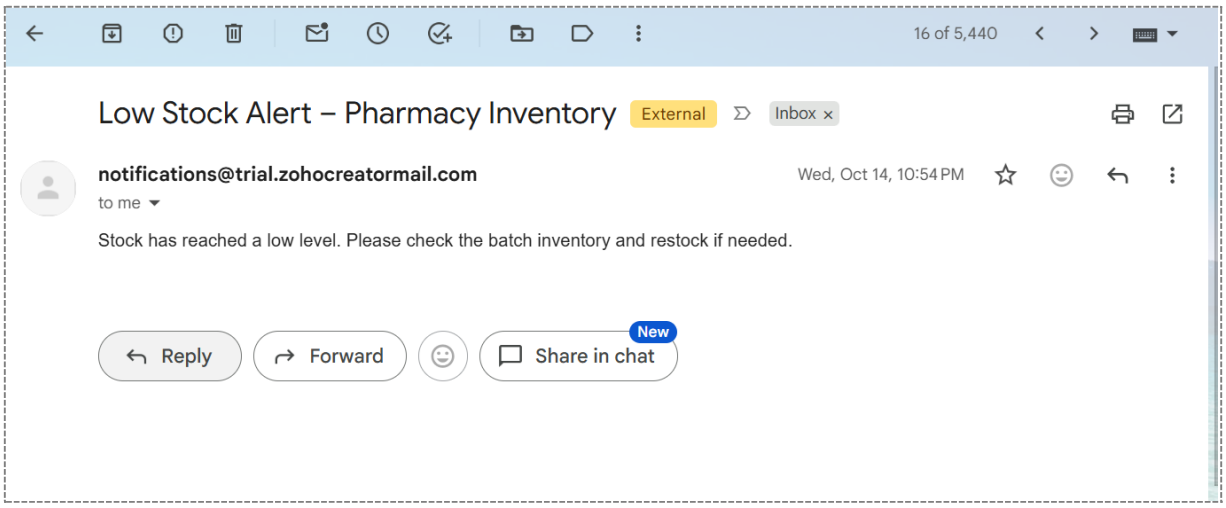


Figure 5.13: Low Stock Alert E-mail

5.8 User Management and Role-Based Access

Access was configured in the Users and Permissions section of Zoho Creator. Three custom permission profiles were created in addition to the default Administrator, Read, Write, Developer and Customer profiles:

- **Pharmacist Access:** allows creating and viewing medicines, suppliers and batch inventory, and creating, viewing and editing stock issues and reorder requests.
- **Inventory Staff Access:** allows staff to monitor and edit batch inventory, record stock issues and submit reorder requests, and to view medicines and suppliers.
- **Pharmacy Admin Access:** allows managing all medicine, supplier, purchase, batch inventory, stock issue, reorder and stock adjustment records, and gives access to the Pharmacy_Dashboard.

A role hierarchy was defined with CEO at the top, Pharmacist below CEO, Inventory Staff below Pharmacist, and Pharmacy Admin also below CEO. A sharing rule named "Batch Inventory Sharing" shares Batch Inventory records from the Inventory Staff role (and subordinates) to the Pharmacist role with read access, so pharmacists can check batch and expiry details entered by inventory staff.

5.9 AI and Analytics Features

Analytics: The dashboard pages provide four KPI cards and charts of stock by medicine, batch status split and available quantity by batch. These give pharmacy staff an immediate overview of stock and pending reorders.

AI features: Zia features were not used in the delivered application. Demand forecasting and prediction of stock-outs from issue history are identified as possible future enhancements.

5.10 Pro-Code Extension

No separate pro-code extension was implemented. The expiry validation, stock update and notifications were configured using Zoho Creator's built-in features and Deluge scripting. Possible future pro-code extensions include a JavaScript widget for barcode scanning of batch numbers, a Zoho Catalyst function for demand forecasting, and integration through the Creator REST API with an accounting or billing system.

5.11 Mobile and Portal Access

The application is available on the Zoho Creator mobile app from the same configuration, so staff can record stock issues and check batches at the shelf. No mobile-specific layout or customer portal was configured.

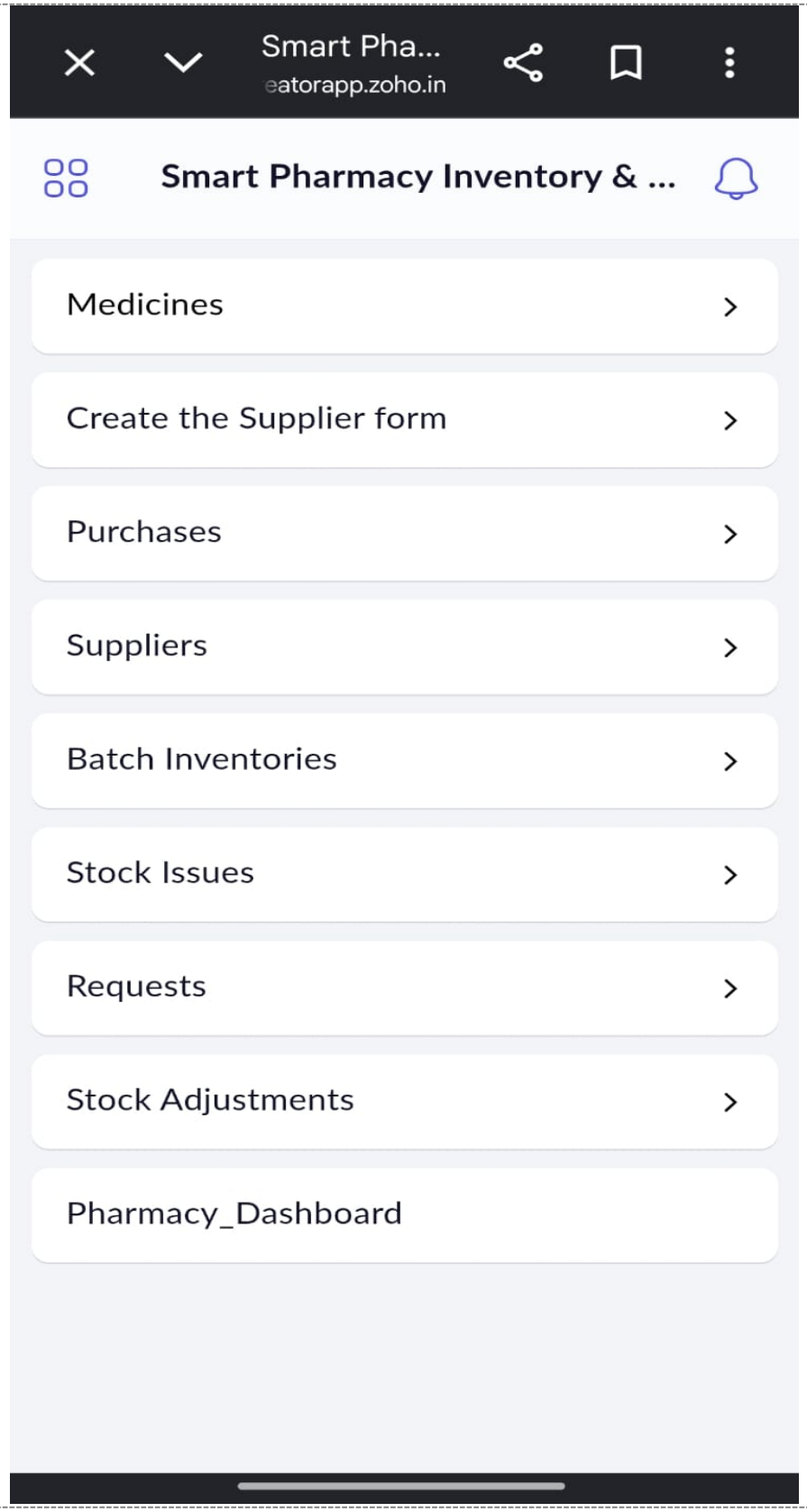


Figure 5.16: Application on the Zoho Creator Mobile App

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing is carried out at four levels:

- Unit testing of each form and script with sample records for medicines, suppliers, purchases and batches.
- Integration testing of the expiry validation, automatic stock update, low-stock alert and reorder notification workflows.
- Access testing by sharing the application with users in each permission profile and checking their access.
- Review of the dashboard KPI values and charts against the stored records.

Each test case is traced to a functional requirement of Table 3.2. The actual result and status columns of Table 6.1 are to be recorded by the student after executing each test in the live application; open issues are listed in the defect and improvement log (Table 6.2).

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result
TC-01	FR-01	Add a medicine	Fill all Medicine fields and submit	Record saved with auto-generated Medicine ID
TC-02	FR-02	Add a supplier	Fill Create the Supplier form and submit	Record saved; supplier appears in the report
TC-03	FR-03	Record a purchase	Select medicine and supplier, enter invoice, quantity and cost	Purchase saved and listed in All Purchases
TC-04	FR-04	Add a batch with a future expiry date	Select medicine and purchase, enter batch number, dates and quantity	Batch saved with success message
TC-05	FR-05	Add a batch with an expiry date in the past	Enter expiry date earlier than today and submit	Alert “Expiry date cannot be in the past.”; submission cancelled
TC-06	FR-06	Record a stock issue	Select batch, quantity, issue type, issued to/by	Stock issue saved
TC-07	FR-07	Automatic stock update (sufficient stock)	Issue quantity less than available quantity	Batch available quantity reduced by the issued quantity
TC-08	FR-07	Issue more than available stock	Issue quantity greater than available quantity	Quantity not reduced; message “Insufficient stock available.”
TC-09	FR-08	Low stock alert	Issue stock so that available quantity is 10 or fewer	Low Stock Alert e-mail received
TC-10	FR-09	Reorder request notification	Submit a new reorder request	Medicine Reorder Request e-mail received

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result
TC-11	FR-10	Urgent Reorders report	Create Pending requests with High/Urgent priority; open report	Requests listed, sorted by priority and date
TC-12	FR-11	Record a stock adjustment	Select batch, type, quantity, reason and approval status	Adjustment saved in All Stock Adjustments
TC-13	FR-12	List reports	Open each of the report in the menu	Records of the respective form displayed
TC-14	FR-13	Dashboard KPIs	Open Pharmacy Dashboard 2	KPI values match the stored records
TC-15	FR-13	Dashboard charts	Open Stock by Medicine and Batch Status Split charts	Charts reflect available quantity and batch status
TC-16	FR-14	Profile access	Sign in as Pharmacist, Inventory Staff and Pharmacy Admin	Forms and reports available as per Table 4.10
TC-17	FR-14	Batch sharing rule	Pharmacist opens batch records added by Inventory Staff	Batch records visible with read access

6.3 Debugging Deluge Scripts

Scripts are debugged with info statements (for example, the “Stock quantity updated.” and “Insufficient stock available.” messages in DS-02), with the alert messages shown on form submission (DS-01), with the workflow list to see which rules are enabled for each form, and by checking the batch’s Available Quantity after each stock issue. Null checks are placed inside the scripts before records are fetched, which avoids errors when a lookup or date field is empty.

6.4 Defect Log

A review of the application configuration identified the following defects and improvements. They are recorded honestly as open items for the next version of the application.

Table 6.2: Defect and Improvement Log

Bug ID	Description	Root Cause	Planned Fix	Status
BUG-01	Stock Adjustment form shows extra Approval Status and Drop Down fields with Choice 1–3	Placeholder fields left from form design	Delete the unused fields and keep Approval Status (Pending / Approved / Rejected)	Open
BUG-02	Two supplier forms exist: Create the Supplier form and Supplier	Second form created for the Reorder Request lookup	Use one supplier form for Purchase and Reorder Request	Open
BUG-03	Urgent Reorders may list urgent requests of any status	Criteria combine && and ‖ without brackets	Rewrite as Request_Status == “Pending” && (Priority == “High” ‖ Priority == “Urgent”)	Open
BUG-04	A stock issue larger than available stock is saved; only an information message is shown	DS-02 runs on success and cannot cancel the submission	Move the quantity check to on validate and cancel submit	Open

Bug ID	Description	Root Cause	Planned Fix	Status
BUG-05	Low-stock threshold (10 units) and recipient e-mail are fixed in the workflow	Values typed directly in the workflow	Compare with Minimum Stock Level / Reorder Level; use a configurable recipient	Open
BUG-06	Expiry check runs only on add, and Batch Status is not set to Expired or Depleted automatically	Workflow event limited to add; no scheduled workflow	Validate on edit as well; add a scheduled near-expiry and expired-batch workflow	Open

6.5 User Acceptance Testing

User acceptance testing is carried out by asking a user of each profile to perform the tasks below and recording the feedback.

Table 6.3: User Acceptance Feedback

User	Role Tested	Tasks Verified	Feedback / Action Taken
Name 1	Pharmacy Admin	Add medicines, suppliers, purchases and batches; view dashboard and Urgent Reorders	Feedback 1
Name 2	Inventory Staff	Update batches, record stock issues, submit reorder requests	Feedback 2
Name 3	Pharmacist	View medicines and shared batch records, record stock issues	Feedback 3

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application was developed, tested and demonstrated within a single Zoho Creator application workspace using the free trial edition. A separate Stage or Production environment was not configured, and the application is used for demonstration purposes. Changes were managed by testing individual forms, workflows and reports after each configuration, and sample records were identified so that they can be removed before real use. A separate Development–Stage–Production deployment process was not implemented in this version.

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	Test cases executed in the application workspace	✓
2	Sample/test data identified	✓
3	Permission profiles and roles configured	✓
4	Connections authorised (none required)	N/A
5	Form workflows enabled	✓
6	Application URL verified	✓
7	Open defects recorded (Table 6.2)	✓

7.2 Performance Optimisation

With a small demonstration dataset, performance was not a major constraint. Basic optimisation practices were followed to keep the application manageable and responsive, as summarised in Table 7.2. Detailed timings were not measured in this version.

Table 7.2: Performance Optimisation Measures

Optimisation Applied	Component	Before	After
Condition on the low-stock alert so that e-mail is sent only at or below the threshold	DS-03	Mail for every stock issue	Mail only when quantity ≤ 10
Null and quantity checks before fetching the batch record	DS-02	Possible error on empty values	Script runs only for valid input
Aggregate KPI cards (count and sum) instead of opening several reports	Dashboard	Multiple reports needed	Summary at a glance
Custom filtered and sorted Urgent Reorders report	Report	Scan all requests	Only urgent requests shown

7.3 Security Features

Security was configured using Zoho Creator’s permission profiles, roles, sharing rules and personal-data settings.

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Permission profiles (Pharmacist, Inventory Staff, Pharmacy Admin)	Forms, reports and dashboard	Unauthorised access or changes
Role hierarchy (CEO, Pharmacist, Inventory Staff, Pharmacy Admin)	Record visibility	Excess data visibility
Sharing rule with read-only access	Batch Inventory form	Unintended edits by pharmacists
Personal data marking	Supplier phone and e-mail fields	Exposure of personal data
No hard-coded credentials; sendmail from zoho.adminuserid	DS-03, DS-04	Credential leakage
Expiry validation and quantity check	DS-01, DS-02	Entry of expired stock; wrong stock quantity

7.4 Governance and Administration

- **Role-based governance:** the Administrator manages users, profiles and roles; Pharmacy Admin manages business data.
- **Naming conventions:** forms are named after business entities and fields use clear link names; workflows are named after their purpose.
- **Test data control:** sample records are identified and are to be removed before live use.
- **Maintenance:** the administrator reviews the workflow list after changes to avoid duplicate rules, and reviews the defect log (Table 6.2) before each release.

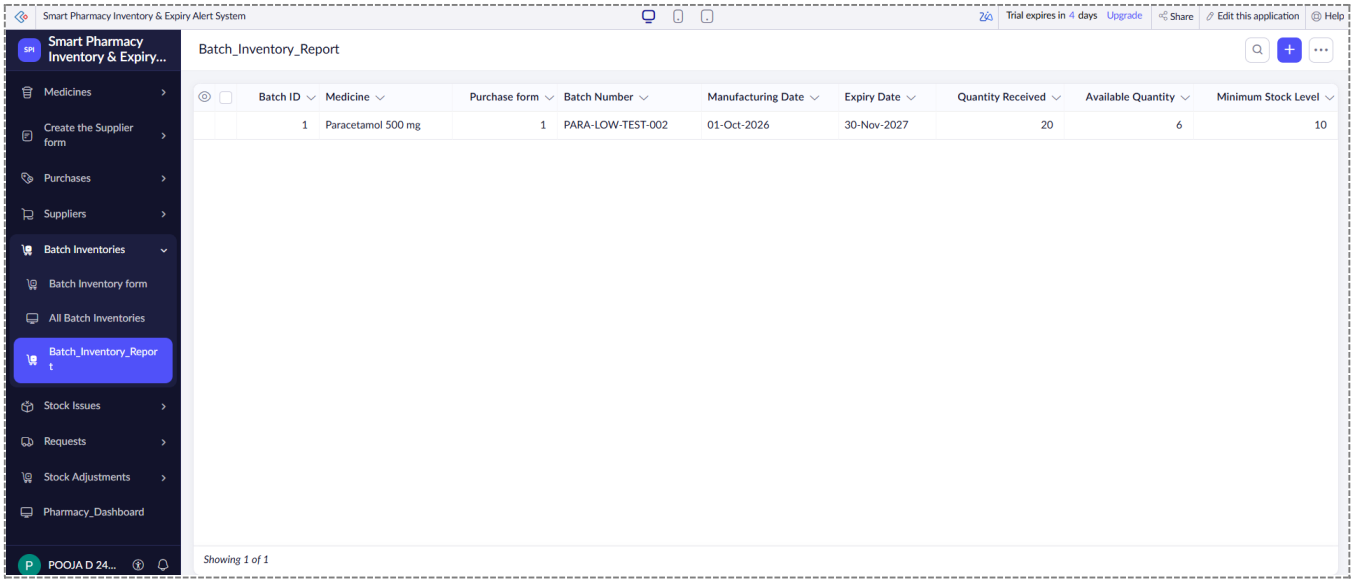


Figure 7.1: Roles and Sharing Rule Configuration

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The application is demonstrated with a scenario that follows a medicine from purchase to reorder. First, a medicine and a supplier are added, a purchase is recorded and the received stock is entered as a batch with its expiry date (Figure 8.1); a batch with an expiry date in the past is rejected. Next, stock is issued from the batch and its available quantity is reduced automatically (Figure 8.2). When the available quantity falls to 10 or fewer units, the Low Stock Alert e-mail is sent, a reorder request is raised and the notification e-mail is received, and the dashboard shows the updated KPI values and charts (Figure 8.3).

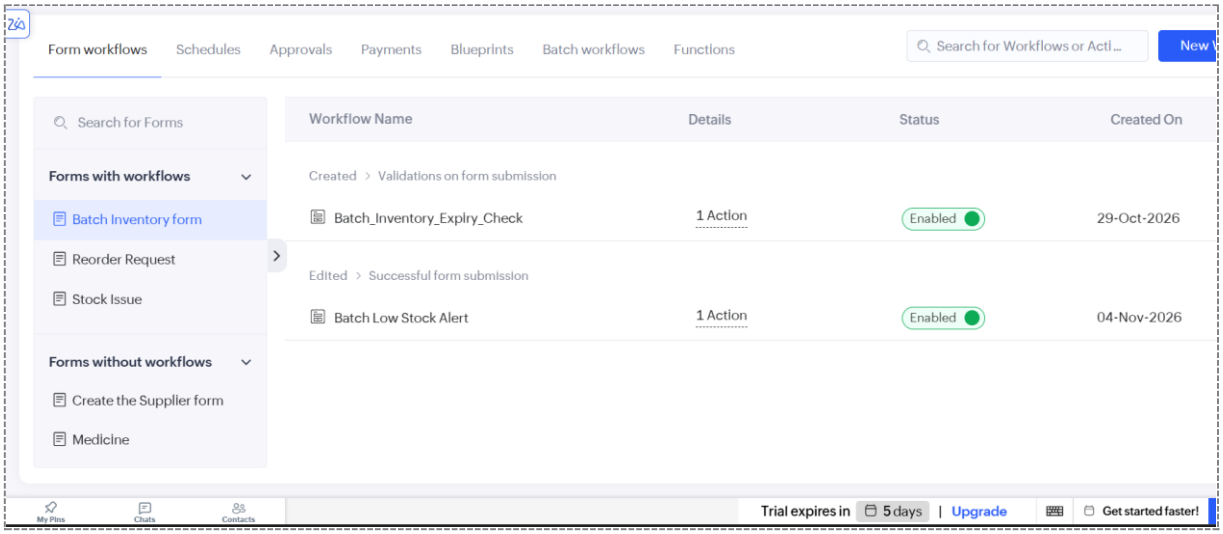


Figure 8.1: Scenario Step 1 – Medicine, Purchase and Batch Added

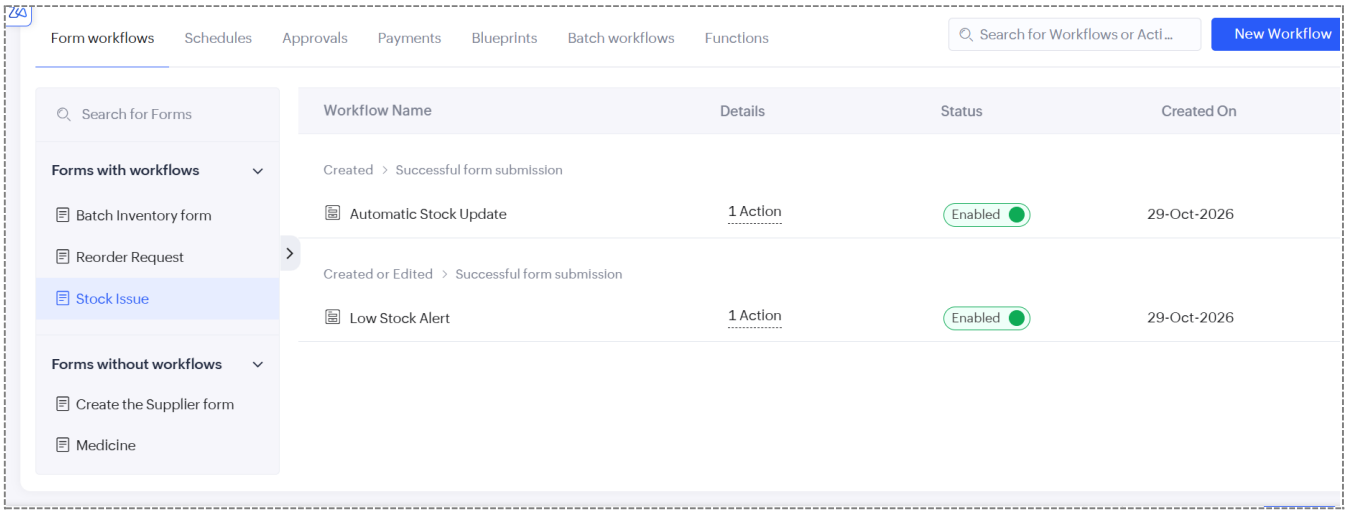


Figure 8.2: Scenario Step 2 – Stock Issue and Automatic Quantity Update

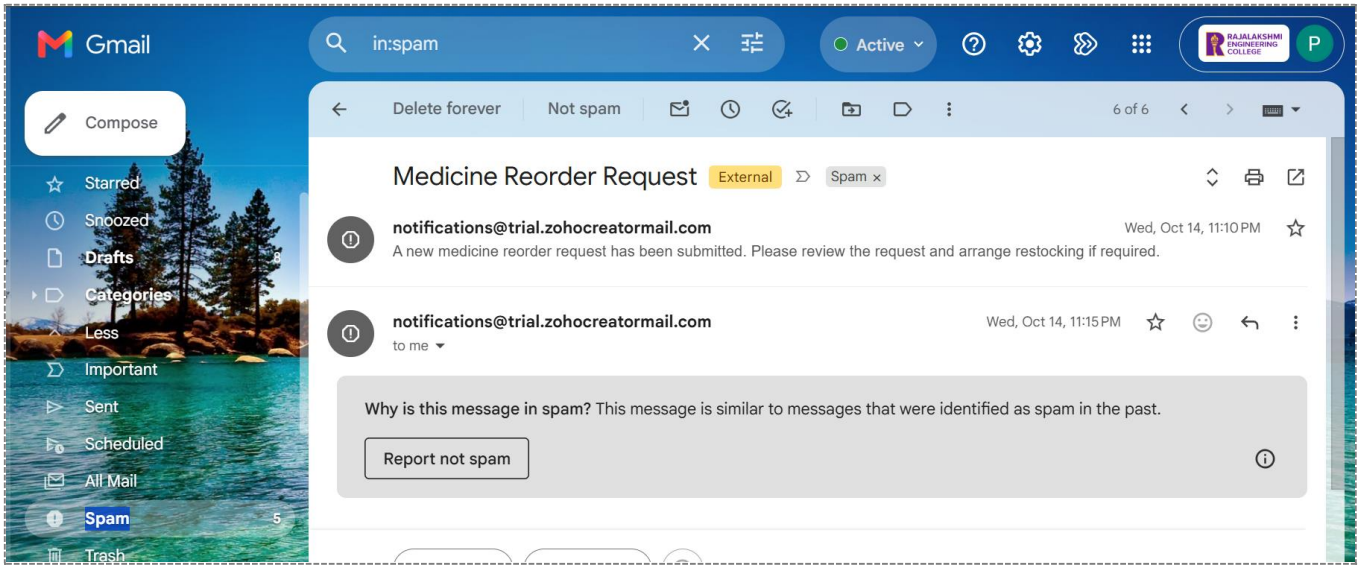


Figure 8.3: Scenario Step 3 – Alerts, Reorder Request and Dashboard

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Design the data model and forms	Sections 4.2, 4.3, 5.2	Eight forms with lookup relationships (TC-01 to TC-04, TC-06, TC-10, TC-12)
Prevent expired batches and update stock automatically	Sections 5.5, 5.6 (DS-01, DS-02)	Expiry validation and automatic quantity update (TC-05, TC-07, TC-08)
Send low-stock and reorder e-mails	Sections 5.6, 5.7 (DS-03, DS-04)	E-mail notifications configured (TC-09, TC-10)
Provide reports and dashboard	Sections 5.3, 5.4	Ten list reports, two dashboard pages (TC-11, TC-13 to TC-15)
Configure access control and governance	Sections 5.8, 7.1–7.4	Profiles, roles and sharing rule (TC-16, TC-17)

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Expiry control	Visual inspection	Past expiry dates rejected at entry; batch status tracked	Safer stock
Stock update	Manual, delayed	Available quantity reduced on issue	Accurate stock
Low-stock alert	Manual counting	E-mail at 10 units or fewer	Early action
Reorders	Verbal or phone	Reorder form with priority, status and notification	Traceable requests
Records	Registers and sheets	Linked forms and reports	Single source of truth
Management view	Collected from registers	Dashboard KPIs and charts	Current overview

8.4 Limitations of the Low-Code Approach

- **Workflow ordering and conditions:** the low-stock alert and the automatic stock update both run on stock issue, so their order and the use of fixed thresholds need careful review.
- **Validation on success events:** a script that runs on success cannot cancel a submission, so insufficient stock is only reported with a message (BUG-04); the check should be moved to the validate event.
- **Not yet implemented:** scheduled near-expiry alerts, automatic Batch Status changes, approval workflows for stock adjustments and reorders, and Blueprints.
- **Plan limits:** the free trial edition does not provide separate environments, and e-mail and execution limits apply.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of the application is no-code: the eight forms, lookups, list reports, dashboard pages, permission profiles, roles and the sharing rule were configured visually. The low-code part consists of the Deluge scripts for expiry validation and stock update and the send-mail notifications. Barcode scanning, demand forecasting and exchange of data with billing systems would require pro-code (widgets, Catalyst or REST APIs).

The project showed that low-code delivers an inventory application quickly, but that careful testing of workflow events, conditions and data values is still essential; moving to pro-code is justified only when a requirement exceeds the platform’s components or must connect to external systems.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Smart Pharmacy Inventory & Expiry Alert System was built on Zoho Creator to manage medicines, suppliers, purchases, batch inventory, stock issues, reorder requests and stock adjustments. It provides eight forms, ten list reports, two dashboard pages, four form workflows with Deluge scripts for expiry validation and automatic stock update, e-mail alerts for low stock and reorder requests through Zoho Mail, and permission profiles, roles and a sharing rule for role-based access.

The application replaces scattered registers with a single, connected record of medicines and batches, prevents expired batches from entering stock, keeps available quantities current and gives the pharmacy early warning of shortages. The defects and improvements identified in Table 6.2 define the work for the next version.

9.2 Future Enhancements

- **Scheduled expiry alerts:** a daily scheduled workflow that e-mails batches expiring within 30, 60 or 90 days and automatically sets Batch Status to Expired or Depleted.
- **Stronger validation:** block stock issues larger than available stock, use the Minimum Stock Level or Reorder Level instead of a fixed threshold of 10, and send alerts to a configurable recipient list.
- **Approvals and Blueprints:** approval processes for stock adjustments and reorder requests, and a Blueprint for the reorder life cycle (Pending → Approved → Ordered → Received).
- **Clean-up:** merge the two supplier forms and remove placeholder fields from the Stock Adjustment form.
- **Pro-code extensions:** barcode scanning of batch numbers, demand forecasting with Zoho Catalyst or Zia, and Creator REST API integration with billing or accounting software.

9.3 Learning Outcomes and Career Relevance

Through this project I learned to analyse a pharmacy inventory process, model related entities with lookups, write Deluge scripts for validation and stock updates, configure e-mail notifications, build dashboards and set up permission profiles, roles and sharing rules. I also learned that workflow events and conditions must be tested end to end and that issues should be documented honestly.

These skills map to roles such as citizen developer and low-code developer, solutions consultant and automation engineer, which combine process understanding with platform configuration and selective scripting.

REFERENCES

[1] Zoho Corporation, “Zoho Creator Help Documentation.” [Online]. Available: https://www.zoho.com/creator/help/. (Accessed: Nov. 2026).

[2] Gartner, Inc., “Magic Quadrant for Enterprise Low-Code Application Platforms,” Oct. 16, 2024. [Online]. Available: https://www.gartner.com/en/documents/5844247.

[3] A. Sahay, A. Indamutsa, D. Di Ruscio, and A. Pierantonio, “Supporting the understanding and comparison of low-code development platforms,” in Proc. 46th Euromicro Conf. Software Engineering and Advanced Applications (SEAA), 2020, pp. 171–178, doi: 10.1109/SEAA51224.2020.00036.

[4] R. Sanchis, Ó. García-Perales, F. Fraile, and R. Poler, “Low-code as enabler of digital transformation in manufacturing industry,” Applied Sciences, vol. 10, no. 1, Art. no. 12, 2020, doi: 10.3390/app10010012.

[5] M. Hirzel, “Low-code programming models,” Communications of the ACM, vol. 66, no. 10, pp. 76–85, 2023, doi: 10.1145/3587691.

[6] Government of India, “The Drugs and Cosmetics Act, 1940 and the Drugs and Cosmetics Rules, 1945.”

[7] World Health Organization, “Model guidance for the storage and transport of time- and temperature-sensitive pharmaceutical products,” in WHO Expert Committee on Specifications for Pharmaceutical Preparations, Technical Report Series No. 961, Annex 9, 2011.

[8] Zoho Corporation, “Deluge Script Reference Guide.” [Online]. Available: https://www.zoho.com/deluge/. (Accessed: Nov. 2026).

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Table No.
I	Application created & sections/menu organised	Y	Smart Pharmacy Inventory & Expiry Alert System with menu sections	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Y	8 forms: Auto Number, Text, Multi Line, Number, Date, Currency, Drop Down, Lookup, Email, Phone, Address	5.2	Tables 4.2–4.9
II	Lookup field (relationship between forms)	Y	Purchase, Batch Inventory, Stock Issue, Stock Adjustment, Reorder Request	4.2, 5.2	Fig. 4.2; Table 4.1
II	Subform (parent–child data)	N	— (not required)	5.2	—
II	Field validations / layout rules	Y	Expiry date validation (DS-01)	5.5	Code 5.1
II	List report with filter, grouping, custom action	Y	Urgent Reorders (filter and sort); 10 list reports	5.3	Table 5.1
II	Kanban / Calendar / Timeline / Map report	N	— (not configured)	5.3	—
II	Pivot table / chart report	N	— (charts are placed on dashboard pages)	5.4	—
II	Page / dashboard built with UI Builder	Y	Pharmacy_Dashboard, Pharmacy Dashboard 2	5.4	Fig. 5.11
II	Deluge – form events (validate / success)	Y	DS-01 to DS-04	5.5	Code 5.1–5.3
II	Deluge – functions, fetch, loops, collections	Y	Record fetch in DS-02 (no stored functions or loops)	5.5	Code 5.2
II	Email / in-app notifications	Y	Low Stock Alert, Reorder Request e-mails	5.6, 5.7	Fig. 5.13
II	Form workflow	Y	Four form workflows	5.6	Table 5.3
II	Scheduled workflow	N	— (planned)	9.2	—
II	Approval process	N	— (planned)	9.2	—
II	Blueprint	N	— (planned)	9.2	—
II	Batch / report workflow	N	— (not required)	5.6	—
II	Testing & debugging evidence	Y	17 test cases; defect log	6	Tables 6.1, 6.2
III	Integration with a Zoho service	Y	Zoho Mail via sendmail	5.7	Table 5.2
III	Connection / REST API call (invokeurl)	N	—	5.7	—
III	Webhook or Creator REST API	N	—	5.7	—
III	Users, roles and permission profiles	Y	Three custom profiles, role hierarchy, sharing rule	5.8	Fig. 5.14; Table 4.10
III	Record-level access criteria / customer portal	N	— (sharing rule only)	5.8	—
III	Zia / AI feature	N	— (planned)	5.9	—
III	Analytics (charts / KPIs)	Y	Dashboard KPIs and charts	5.9	Fig. 5.15
IV	Performance optimisation applied & measured	Y	Measures applied; timings not measured	7.2	Table 7.2
IV	Deployment across environments	N	Single workspace (free trial)	7.1	Table 7.1
IV	Security controls (permissions, personal data)	Y	Profiles, roles, sharing rule, personal data fields	7.3	Table 7.3

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Table No.
IV	Governance (administration, test-data control)	Y	Administrator review, naming conventions	7.4	Fig. 7.1
V	Pro-code extension (Catalyst / widgets / functions)	N	— (justified in 5.10)	5.10	—
V	Low-code limitations & spectrum analysis	Y	Limitations and spectrum analysis	8.4, 8.5	—
—	Mobile app / portal access	Y	Zoho Creator mobile app (not separately tested)	5.11	Fig. 5.16

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2		
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6		
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9		
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7		
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3		

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Batch_Inventory_form – Expiry Date Validation (DS-01)

```
// DS-01 | Form: Batch_Inventory_form | Event: On add - On validate
if(input.Expiry_Date != null && input.Expiry_Date < zoho.currentdate)
{
    alert "Expiry date cannot be in the past.";
    cancel submit;
}
```

C.2 Stock_Issue – Automatic Stock Update (DS-02)

```
// DS-02 | Form: Stock_Issue | Event: On add - On success
if(input.Batch_Inventory_form != null && input.Quantity_Issued > 0)
{
    batchRecord = Batch_Inventory_form[ID == input.Batch_Inventory_form];
    if(batchRecord.count() > 0)
    {
        currentQty = batchRecord.Available_Quantity;
        issuedQty = input.Quantity_Issued;
        if(currentQty >= issuedQty)
        {
            batchRecord.Available_Quantity = currentQty - issuedQty;
            info "Stock quantity updated.";
        }
        else
        {
            info "Insufficient stock available.";
        }
    }
}
```

C.3 Stock_Issue – Low Stock Alert (DS-03)

```
// DS-03 | Form: Stock_Issue | Event: On add or edit - On success
// Criteria: Batch_Inventory_form.Available_Quantity <= 10
sendmail
[
    from : zoho.adminuserid
    to : "pooja.d.2024.cse@rajalakshmi.edu.in"
    subject : "Low Stock Alert – Pharmacy Inventory"
    message : "Warning: Available medicine stock has reached 10 or fewer units.
    Please arrange restocking."
]
```

C.4 Reorder_Request – Reorder Notification (DS-04)

```
// DS-04 | Form: Reorder_Request | Event: On add - On success
sendmail
[
    from : zoho.adminuserid
    to : "pooja.d.2024.cse@rajalakshmi.edu.in"
    subject : "Medicine Reorder Request"
    message : "A new medicine reorder request has been submitted. Please review the
    request and arrange restocking if required."
]
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Web: open the application link and sign in with the Zoho account that the application owner shared the app with. Mobile: install the Zoho Creator app, sign in with the same account and open the application. The forms and reports available depend on your permission profile.

D.2 Instructions for the Pharmacy Admin

- Add medicines in the Medicine form and suppliers in the Create the Supplier form.
- Record each delivery in the Purchase form, then add its batch in the Batch Inventory form with the batch number, dates, quantity and expiry date (past expiry dates are rejected).
- Review Requests and Urgent Reorders and update the request status; record write-offs in Stock Adjustment.
- Open Pharmacy_Dashboard to monitor total medicines, suppliers, stock and pending reorders.

D.3 Instructions for the Inventory Staff

- Open Batch Inventories to monitor and update batch quantities, status and storage location.
- Record stock issues in Stock Issue; the batch's available quantity is reduced automatically.
- Submit a Reorder Request with priority and suggested quantity when stock is low.

D.4 Instructions for the Pharmacist

- View medicines, expiry dates and the batch records shared by Inventory Staff.
- Record medicines issued in Stock Issue and raise a Reorder Request when needed.

D.5 Instructions for the Administrator

- Add users, assign permission profiles and roles, and review the workflow list after any change.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	POOJA D	**Register No.**	240701385
Project Title	SMART PHARMACY INVENTORY & EXPIRY ALERT SYSTEM	**Date**	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner Name: _____	Signature of External Examiner Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	POOJA D	**Register No.**	240701385
Department / Section	CSE / G	**Date of Viva**	
Application URL Verified	Yes / No	**Demo Duration**	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

___

INTERNAL EXAMINER Name: _____	EXTERNAL EXAMINER Name: _____